

The Evolution of Urban - Rural Income and Consumption Inequality in Transitioning China *

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15th April 2026

Abstract

Despite China's extraordinary economic growth, both income and consumption inequality among households rose sharply between 1989 and 2009, with income inequality increasing more rapidly. In this paper, we quantify how private household savings have mitigated the consequences of rising income uncertainty on their consumption. An increase in permanent income risk can generate wider dispersion in both incomes and consumption, consistent with the observed rise in inequality. Both rural and urban households showed a strong ability to smooth their consumption against permanent income shocks through personal savings but were far from achieving complete insurance. On the other hand, transitory idiosyncratic income shocks are perfectly and easily insured by saving. Our findings using the Chinese case study add evidence to the strand of literature documenting partial insurance of permanent shocks and full insurance of transitory shocks.

Keywords: Income Inequality, Consumption Inequality, Income Shocks, China

JEL Classification Numbers: C61, D15, E21, G51

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*We thank Areendam Chanda, Faik Koray, Gary C. Sanger, Leng Ling, Chunzan Wu, and seminar participants at 2024 Southern Economic Association Meetings, 2025 CES North America Conference, 2025 Spring Midwest Macro Meetings, and 2025 WEAI Annual Conference for insightful comments and suggestions. Disclaimer: The views expressed in this paper are solely those of the authors and do not necessarily reflect the views of the Federal Reserve Bank of Dallas or the Federal Reserve System. Any errors or omissions are the responsibility of the authors.

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1 Introduction

The rapid growth of the Chinese economy has drawn wide academic attention (Song et al., 2011; Zhu, 2012; Brandt and Rawski, 2008). However, little research has analyzed the welfare consequences of massive economic reforms imposed on Chinese households, with Santaaulalia-Llopis and Zheng (2018) being a notable exception. Following Blundell and Preston (1998), they estimate the transmission of permanent and transitory labor income shocks to consumption, and analyze the welfare effects of economic growth. While Santaaulalia-Llopis and Zheng (2018)’s approach is silent about the exact mechanisms of how households share risk, we adopt a tractable buffer-stock/life-cycle framework promoted by Carroll (1997) and Gourinchas and Parker (2002) to measure the degree of consumption smoothing achieved through a particular channel: personal saving.

This paper focuses on the income and consumption inequalities found in Chinese microdata for the period 1989–2009. We find that both labor income inequality and consumption inequality have increased remarkably, with the increase in income inequality outpacing that of consumption inequality. We attempt to investigate the following questions: How much does the increase in permanent and transitory shocks to Chinese household incomes contribute to changes in income inequality? Are Chinese households able to secure their consumption against increasingly uncertain labor income through their self-insurance mechanism of private saving?

An unprecedented increase in the Chinese household savings rate motivates this research, which has driven increases in the aggregate savings rate, and in turn fueled low-return investment and external imbalances (Wei and Zhang, 2011; Chamon et al., 2013; Curtis et al., 2015; Zhang et al., 2018; Dotsey et al., forthcoming; Yang, 2023). In this paper, we are interested in how households’ self-insurance through savings has mitigated the consequences of the rising labor income uncertainty on their consumption. In the model, we assume that households save and borrow through a risk-free asset to allocate consumption across time periods. We capture other risk-sharing channels such as flexible labor supply, social insurance programs, etc. by using disposable income directly from the data.¹

In our buffer-stock life-cycle model, agents are assumed to face income shocks that contain a persistent component and a transitory one. Households then make consump-

¹See Heathcote et al. (2014) and Pijoan-Mas (2006) for similar practices.

tion/saving decisions at each age to maximize their discounted stream of lifetime utility. The only consumption smoothing mechanism is self-insurance through saving, given the knowledge of income growth rates and the degree of income uncertainty. Savings earn a risk-free interest rate. There is a pay-as-you-go social security system for households provided by the government, in which payment is tied to permanent income, thus providing partial insurance against transitory income shocks.

An important aspect of our analysis is that we attempt to shed light on the different levels of income shocks experienced by rural and urban households. Empirically, we find that rural households displayed greater income dispersion than urban households in all waves. However, the increase in income inequality was less steep in rural households than in urban households between 1993 and 2009: the variance of the logarithm of earnings increased from 0.89 to 1.25 for rural households and from 0.45 to 0.85 for urban households. This motivates us to estimate the permanent and transitory income risks for rural and urban households separately in order to characterize differences in their income processes during the life cycle.

We next empirically measure consumption insurance between rural and urban China. Consumption is much more equal than income in both rural and urban samples, providing evidence to consumption-smoothing behaviors among Chinese households. Consumption inequalities, measured as the variance of the logarithm of consumption, increased in both rural and urban samples from 1993 to 2009, with the rural sample experiencing a significantly larger increase than the urban sample. Consumption inequality increased by nearly 0.20 from 0.24 to 0.43 for rural households and by 0.12 from 0.19 to 0.31 for urban households.

In our model, we explicitly account for a few notable differences in rural and urban households. First, we let rural agents start their life cycles with a lower level of initial permanent income than urban households, consistent with evidence in our empirical data. Second, rural households face greater permanent and transitory income shocks along the life cycle than urban households. Third, our model also features distinctive age-specific survival rates between urban and rural and income growth rates and captures the fact that rural households have a lower life expectancy than urban households. Then we separately simulate rural and urban households.

The results of our model simulation led us to conclude that the increase in permanent

income variance is the main contributor to the rise in income inequality. Income fluctuations were partially insured through personal saving, which led to a smaller increase in consumption inequality. We found evidence of partial insurance against permanent income variations and perfect insurance against transitory income variations. Surprisingly, our rural households showed a strong ability to self-insure against income risk.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data and key empirical facts. Section 4 outlines the model, and Section 5 details the calibration procedure. Sections 6 and 7 compare the model's results with empirical evidence for 1993 and over time, respectively. Section 8 presents a sensitivity analysis, and Section 9 concludes.

2 Related Literature

Our paper is related to the body of literature on consumption inequality. [Krueger and Perri \(2006\)](#), [Blundell et al. \(2008\)](#), [Primiceri and Van Rens \(2009\)](#), and [Heathcote et al. \(2010\)](#) are among the first papers to document the divergence between the U.S. income inequality and consumption inequality in the cross section. [Krueger and Perri \(2006\)](#) attribute this divergence to the development of the credit market. [Blundell et al. \(2008\)](#) find that the shift in the persistence of income shocks can explain this divergence.² [Storesletten et al. \(2004\)](#) examines risk sharing along the life cycle, finding that uninsurable earnings shocks can account for rising income and consumption inequality over the life cycle. [Heathcote et al. \(2014\)](#) quantify risk sharing within an incomplete market model with partial insurance.³ [Aguilar and Bils \(2015\)](#) and [Fisher et al. \(2013\)](#), however, find a much closer co-movement of consumption inequality and income inequality using alternative measures of consumption.

Ample papers study the link between income inequality and consumption inequality in other developed countries. [Barrett et al. \(2000\)](#) find more equal consumption than income in Australia. [Jappelli and Pistaferri \(2010\)](#) document that income inequality in Italy is both higher and has risen more rapidly than consumption inequality. [Brzozowski et al. \(2010\)](#) come to a similar conclusion to [Jappelli and Pistaferri \(2010\)](#) using data from

²[Kaplan and Violante \(2010\)](#) assess the degree of risk sharing using a calibrated life cycle version of the standard incomplete market model and compare the model results with the empirical estimates of [Blundell et al. \(2008\)](#) on the U.S. data.

³Refer to [Attanasio and Weber \(2010\)](#) for a survey of the literature on life-cycle models of consumption.

Canada. [Jappelli and Pistaferri \(2014\)](#) estimate a 0.48 marginal propensity to consume out of transitory shocks using Italian household survey data. [Fella et al. \(2016\)](#) estimate a buffer-stock saving model to quantify the degree of insurance against income risks in Italy, finding that Italian households were able to insure against almost all transitory income shocks, but very little against permanent income risks.

In contrast to the aforementioned papers, we are interested in the evolution of income inequality and consumption inequality in China, a country that has experienced rapid economic growth and has become the world's second largest economy. Income inequality in China has received a great level of academic attention ([Li et al., 2013](#); [Xie and Zhou, 2014](#); [Yang, 1999](#)). However, little research has been done on consumption inequality, let alone the joint evolution of income inequality and consumption inequality in China. [Santaaulalia-Llopis and Zheng \(2018\)](#) find that the rising income inequality has not been accompanied by an equal rise in consumption inequality in China, and that consumption insurance has deteriorated along the process of Chinese economic growth. [Cai et al. \(2010\)](#) and [Ding and He \(2018\)](#) report stronger co-movement between income inequality and consumption inequality using alternative data. While [Santaaulalia-Llopis and Zheng \(2018\)](#) are mute on the exact channels by which households share risk, we adopt a tractable buffer-stock/life-cycle framework to measure the degree of consumption smoothing achieved through a personal savings channel.

Our paper is also related to the literature on the high household saving rate China has experienced during this period. This strand of literature has attributed the high saving rate to the rising private burden of expenditure on education and health care ([Chamon and Prasad, 2010](#); [Imrohoroglu and Zhao, 2018](#)), an unbalanced sex ratio ([Wei and Zhang, 2011](#)), the one-child policy ([Banerjee et al., 2023](#); [Curtis et al., 2015](#); [Choukhmane et al., 2023](#); [Ge et al., 2018](#)), precautionary savings ([Chamon et al., 2013](#); [He et al., 2018](#)), and the interaction of demographers and industry policies ([Dotsey et al., forthcoming](#)). This literature has mostly focused on the aggregate economy or urban households. Relative to this literature, our paper captures the impact of the labor market environment faced by urban and rural households on consumption and saving.

3 Data and Empirical Facts

We use the China Health and Nutrition Survey (CHNS) from 1989 to 2009 to compute income and consumption inequalities, permanent and transitory income shocks, as well as life-cycle income growth rates.⁴ The survey draws samples randomly from nine provinces and autonomous cities/districts, including Guangxi, Guizhou, Heilongjiang, Henan, Hubei, Hunan, Jiangsu, Liaoning, and Shandong. The survey includes detailed demographic, economic, time use, labor force participation, and expenditure information, making it possible to compile consumption and income data at the household level.⁵ In our analysis sample, we keep households with heads aged between 25 and 60 since the number of observations tends to decrease quickly after age 60. We select households between two and six members to control for heterogeneity in household size; ultimately, 2,237 of 25,969 observations were dropped from the sample.

Table 1 presents the summary statistics by wave including the number of observations, the average size of the household, the average age of the head of the household, the disposable income and consumption of the household, as well as the educational attainment of our sample. The surveys sampled around 1,000 urban households and 2,000 rural households per wave, as a result, the rural sample size is double the urban sample size. The average household size shrank slightly from 1989 to 2009 in both rural and urban areas, while the average age of household head increased from 39 (43) to 47 (47) in the rural (urban) sample.

Our benchmark household disposable income consists of various types of earnings, including after-tax labor market income, agricultural income, business income, and public and private transfers. Our benchmark measure of consumption corresponds to expenditures on food, utilities, health services, and semi-durable supplies. Measured in a constant 2009 RMB price, urban households consistently enjoyed higher disposable income than their rural counterparts. Despite the fact that there was a widening income gap between urban and rural households - from around 400 RMB in 1989 to 1,400 RMB in 2009 - average household consumption levels remained comparable, indicating a decline in saving

⁴Data was jointly collected by the Caroline Population Center at the University of North Carolina at Chapel Hill and the National Institute of Nutrition and Food Safety at the Chinese Center for Disease Control and Prevention.

⁵Chinese Household Income Project Series is another nationally representative and publicly available data source that survey household income and consumption. However, it lacks the panel dimension, and the data is sparse as the survey only occurred in 1988, 1995, 2002, 2007, and 2013.

Table 1. Summary Statistics of CHNS, 1989–2009

Wave	Observations (Households)	Household	Head	Disposable				At Least Some	
		Size	Age	Income		Consumption		High School	
		Mean	Mean	Mean	Std. dev.	Mean	Std. dev.	Mean	Std. dev.
Rural									
1989	1,852	4.1	39	1,111	814	1,055	502	0.17	0.37
1991	1,790	4.2	40	1,172	827	1,349	643	0.15	0.36
1993	1,696	4.2	41	1,226	983	1,507	776	0.16	0.36
1997	1,723	4.1	42	1,647	1,307	1,740	1,116	0.17	0.38
2000	1,930	4.0	43	1,962	1,595	1,483	936	0.18	0.38
2004	1,940	3.9	45	2,063	1,793	1,650	952	0.19	0.39
2006	1,823	4.0	45	2,425	2,219	1,441	938	0.23	0.42
2009	1,755	4.0	47	3,590	3,255	1,726	1,103	0.25	0.43
Urban									
1989	885	3.9	43	1,591	798	1,116	559	0.40	0.49
1991	788	4.0	42	1,617	759	1,391	651	0.36	0.48
1993	677	3.9	43	1,758	1,080	1,534	726	0.39	0.49
1997	799	3.7	43	2,022	1,179	1,589	854	0.55	0.49
2000	836	3.5	44	2,627	1,721	1,476	903	0.54	0.50
2004	818	3.4	45	3,065	2,375	1,572	906	0.56	0.49
2006	864	3.5	46	3,422	2,590	1,375	776	0.60	0.49
2009	847	3.5	47	4,880	3,546	1,653	887	0.57	0.49

Notes: This table reports the summary characteristics of the sample used to estimate the labor income and consumption inequality. The statistics include household head age, household size, disposable income and consumption, etc. Household disposable income and consumption are in 2009 RMB.

rates for rural households (see also [Yang, 2023](#)). The educational attainment of household heads improved in both samples, with the share of rural (urban) household heads having at least some high school education rising from 17 (40) percent in 1989 to 25 (57) percent in 2009.

For both rural and urban households, we compute two measures of income and consumption inequality: the cross-sectional variance of the logarithms and the Gini coefficients. To compute income and consumption variance, we first regress the logarithm of disposable income on household characteristics including the sex and age of the household head, household size, education, province of residence, and ethnic minority status. This is done separately by year and for urban and rural households. We then compute the inequality using the residual. This measure of inequality is known as the “within-group” inequality ([Krueger and Perri, 2006](#)). Figure 1 shows a common pattern for rural and urban households: both income inequality and consumption inequality increased from 1989 to 2009.

Noticeably, the income inequality measured by log variance increased from approximately 0.89 to more than 1.25 (increasing over 40%) in the rural sample and from around

0.45 to 0.85 (nearly doubling) in the urban sample over the sample period. The increases in consumption inequality are smaller than the increases in income inequality in both samples. Consumption inequality increased from 0.24 to about 0.43 (almost doubling) in the rural sample and from around 0.19 to nearly 0.31 (over 60%) in the urban sample. The fact that consumption inequality is always less than income inequality provides evidence to some degree of consumption smoothing in the life cycle. However, the fact that consumption inequality increases along with income inequality, although to a much lesser extent, indicates incomplete risk sharing as part of the income shocks managed to pass to consumption.

In Appendix [A.1](#), we show similar variances with and without controls for education and/or gender. Our results remain robust regardless of the inclusion of controls on education or gender, indicating that changes in inequality are not attributed to changing returns related to education or gender. Those observations led us to believe that the increase in income and consumption inequality are largely driven by within-group trends. This is in contrast with [Krueger and Perri \(2006\)](#), who use U.S. data and decompose the inequality to “within group” and “between group” to find comparable “between-group” increases in income and consumption inequality, but the “within-group” income inequality increases remarkably more than the “within-group” consumption inequality. The worsening of consumption inequality across time in Chinese data is also in sharp contrast with that in the U.S. data. For example, [Heathcote et al. \(2023\)](#) find that inequality in consumption expenditures has remained remarkably stable in the United States over the years 1967 - 2021.

Next, we turn to the inequality measured using Gini coefficients. The income Gini coefficients increased from approximately 0.40 (0.28) in 1989 to 0.48 (0.39) in 2009 for the rural (urban) sample. The Gini coefficient on consumption is always less than that on income, and the difference is especially large for the rural sample. The consumption Gini coefficients increased from 0.27 (0.28) in 1989 to 0.35 (0.3) in 2009 in the rural (urban) sample.

Why has income inequality increased dramatically? How much of the increase in income variances is due to the persistent component? How much is due to the transitory one? To answer these questions, we estimate the variances in permanent and transitory income using minimum distance estimation. Appendix [A.2](#) describes the estimation

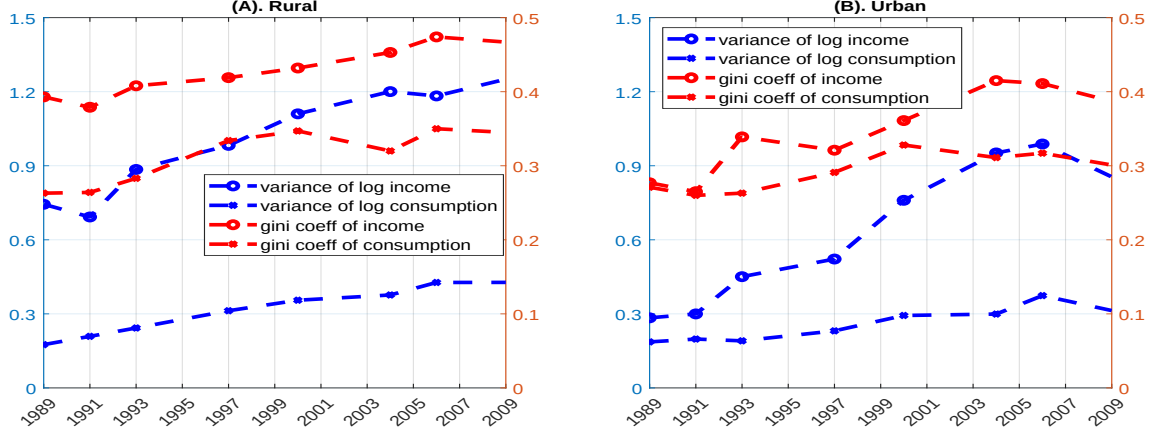


Figure 1. Gini Coefficients and Variance of Logarithm of Income and Consumption for Rural (left) and Urban (right) Households.

process in detail.

To measure the shocks to income, we first regress the logarithm of adult-equivalent income separately for urban and rural households and by year, on a set of dummies on sex of household head, household head age, education, province of residence, and ethnic minority.⁶ The residuals are used to estimate the permanent and transitory variances. In particular, we assume the following process for the residual income. Let y_t be the residual of income at time t , and w_t and v_t be the permanent and transitory components of income, respectively:

$$y_t = w_t + v_t$$

$$w_t = w_{t-1} + \theta_t,$$

where θ_t is the shock to the persistent component of income (permanent shock), $\theta_t \sim \mathcal{N}(0, \sigma_{\theta_t}^2)$, and transitory shocks to income are the transitory component of income, $v_t \sim \mathcal{N}(0, \sigma_{v_t}^2)$. Permanent and transitory shocks are assumed to be mutually orthogonal and i.i.d. across households, time, and age.

The estimated permanent and transitory income variances using disposable income are reported in Table 2. As a robustness check, we also report the results using alternative definitions of income such as earnings plus public transfers, earnings plus private transfers, as well as earnings only. For urban households, permanent income shocks increased from 0.036 in 1992-1993 to 0.076 in 1994-1997, decreased to 0.046 in 1998-2000, increased again

⁶We define adult-equivalent income by dividing household disposable income by household size.

Table 2. Permanent and Transitory Income Risks

	Disposable Income		Earnings + Public Transf.		Earnings + Private Transf.		Earnings Only	
	Rural	Urban	Rural	Urban	Rural	Urban	Rural	Urban
<i>Permanent shocks</i> $\sigma_{\theta_t}^2$								
1992–3	0.099 (0.016)	0.036 (0.013)	0.094 (0.016)	0.040 (0.014)	0.117 (0.016)	0.044 (0.020)	0.108 (0.015)	0.035 (0.015)
1994–7	0.072 (0.019)	0.076 (0.022)	0.059 (0.016)	0.058 (0.015)	0.079 (0.015)	0.087 (0.027)	0.072 (0.015)	0.045 (0.010)
1998–2000	0.072 (0.018)	0.046 (0.024)	0.051 (0.023)	0.051 (0.029)	0.090 (0.023)	0.032 (0.032)	0.071 (0.021)	0.034 (0.014)
2001–4	0.103 (0.018)	0.063 (0.023)	0.085 (0.015)	0.087 (0.028)	0.108 (0.019)	0.172 (0.027)	0.084 (0.013)	0.030 (0.014)
2005–6	0.120 (0.026)	0.113 (0.032)	0.108 (0.024)	0.049 (0.028)	0.142 (0.028)	0.231 (0.046)	0.089 (0.019)	0.045 (0.023)
<i>Transitory shocks</i> $\sigma_{v_t}^2$								
1991	0.296 (0.020)	0.126 (0.018)	0.294 (0.023)	0.126 (0.020)	0.310 (0.024)	0.145 (0.027)	0.306 (0.024)	0.127 (0.021)
1993	0.393 (0.041)	0.219 (0.028)	0.395 (0.048)	0.197 (0.036)	0.360 (0.035)	0.183 (0.040)	0.360 (0.037)	0.162 (0.034)
1997	0.497 (0.054)	0.212 (0.060)	0.453 (0.052)	0.233 (0.053)	0.482 (0.060)	0.289 (0.079)	0.436 (0.052)	0.110 (0.041)
2000	0.501 (0.051)	0.361 (0.062)	0.505 (0.056)	0.416 (0.061)	0.507 (0.054)	0.367 (0.059)	0.411 (0.052)	0.240 (0.037)
2004	0.496 (0.038)	0.326 (0.058)	0.439 (0.039)	0.318 (0.057)	0.458 (0.045)	0.307 (0.063)	0.400 (0.035)	0.253 (0.035)
2006	0.415 (0.042)	0.187 (0.046)	0.408 (0.033)	0.219 (0.049)	0.393 (0.047)	0.219 (0.074)	0.370 (0.040)	0.145 (0.045)
Observations	15,386	6,920	15,357	6,908	15,380	6,903	15,340	6,868

Notes: This table reports the estimation results of the permanent and transitory variances for various adult-equivalent measures of household income. The estimates are obtained for rural and urban samples separately. We assume that permanent and transitory variances remain the same in the years between two consecutive surveys. The standard errors are computed based on 50 bootstrap replicas.

to 0.063 in 2001–2004, then increased further to 0.113 in 2005–2006. Similar patterns can be observed among other urban income measures. For the rural sample, permanent variance of disposable income first decreased from 0.099 in 1992–1993 to 0.072 in 1994–2000, increased to 0.103 in 2001–2004, and further increased to 0.120 in 2005–2006.

In terms of the evolution of transitory risks, both samples experienced an hump-shaped pattern, with peaks in 2000 in both samples. In nearly any given year, the rural sample exhibits both higher permanent and transitory variances than their urban counterpart, which is likely due to the greater risks inherent to agricultural production and the greater price volatility for agricultural products (Santaeulalia-Llopis and Zheng, 2018).⁷

⁷Our observations of income variations are similar to those of Santaeulalia-Llopis and Zheng (2018), who also estimate permanent and transitory income risks using CHNS data using a measurement of income that includes capital income. They also find rapid increases in permanent income risk after 1998 in both rural and urban households, as well as an inverted U-shape evolution of transitory risk between 1989 and 2009. Instead of including capital income in our empirical measure, we consider capital income as endogenous decisions in our quantitative model and exclude it from our benchmark

The observed patterns of permanent and transitory income shocks are a reflection of the larger theme of economic reform and massive structural shifts the Chinese economy experienced in the late 1990s and early 2000s. Urban households, the majority of which were previously employed by state-owned enterprises (SOEs), experienced mass privatization at this time. This change of ownership not only led to higher unemployment and less generous employee benefits for urban residents, but also to higher income uncertainties for rural migrants seeking urban employment.

Those estimates of permanent and transitory variances are the baseline parameters when selecting the permanent variance and transitory variance of income in our quantitative model.

4 The Model

We use a buffer-stock life-cycle model to quantify the impact of increasing income inequality on consumption inequality. There are two stages in an agent's life, a working stage and a retirement stage, and our model period is one year. Agents start working and making consumption/saving decisions at the age of 25 and live with certainty until age 60. After age 60, agents begin facing a survival probability until they die with certainty at age 100. Agents are allowed to borrow and lend across periods but net worth has to be non-negative.⁸

Prior to retirement at age R , agents are assumed to face both permanent and transitory income shocks. Upon their retirement, they start earning a deterministic income which is a percentage of the income they earn at their last working age, determined by the pension replacement ratio.

Household disposable income is denoted by Y_t where t denotes age. During the working stage, we characterize the income process as follows:

measure of household disposable income.

⁸This liquidity constraint has negligible impact on our theoretical income and consumption inequality, as well as saving rate, because when future income is uncertain, the precautionary saving motive implicit in the buffer-stock model naturally discourages risk-averse agents from borrowing from any future income to finance current consumption. See similar discussion in [Choi et al. \(2017\)](#), and theoretical proof in [Yang \(2023\)](#).

$$\begin{aligned}
Y_t &= W_t V_t, \\
W_t &= G_t W_{t-1} \Theta_t, \quad \text{if } t \leq R,
\end{aligned} \tag{1}$$

where W_t represents permanent income, G_t denotes the age-specific gross income growth rate (which is $1 + \text{income growth rate}$), and Θ_t and V_t correspond to the exponential forms of the permanent and transitory income shocks and satisfy $\ln \Theta_t \sim \mathcal{N}(0, \sigma_{\theta_t}^2)$ and $\ln V_t \sim \mathcal{N}(0, \sigma_{v_t}^2)$, respectively. We assume they are mutually orthogonal and i.i.d. across households, age, and time. Agents have perfect foresight of the income shocks in the life cycle.

During the retirement stage, we characterize income process as follows:

$$\begin{aligned}
Y_t &= \eta W_t, \\
W_t &= W_{t-1}, \quad \text{if } t > R,
\end{aligned} \tag{2}$$

where η is the pension replacement ratio.

Households discount the future at rate β . We use s to denote the age-dependent survival rate. Households have preferences over consumption, and the period flow of utility is given by the standard CRRA utility function, $\mu(C_t) = \frac{C_t^{1-\gamma}}{1-\gamma}$, where C_t denotes the current consumption at time t .

Given the life-cycle income process, agents make consumption/saving choices to maximize their expected lifetime utility:

$$\max \left\{ \sum_{t=25}^{100} \beta^t \left(\prod_{j=25}^t s_j \right) E_t[\mu(C_t)] \right\}, \tag{3}$$

$$\text{s.t. } A_{t+1} = (1+r)(A_t + Y_t - C_t), \tag{4}$$

$$A_t \geq 0, \quad \forall t. \tag{5}$$

where A_t is the value of real assets, and r is the annual interest rate.

Define $X_t = A_t + Y_t$ as “cash on hand”, the resource available for consumption at time

period t . Bellman's Principle of Optimality (or recursive form) of the above problem is:

$$V(t, X_t) = \max_{C_t, X_{t+1}} \{\mu(C_t) + \beta s_t V(t+1, X_{t+1})\} \quad (6)$$

$$\text{s.t. } X_{t+1} + C_t = X_t, \quad (7)$$

$$C_t, X_{t+1} \geq 0, \forall t. \quad (8)$$

$V(T+1, X) = 0$, for $t = 1, \dots, T$. The Bellman equation can be simplified to:

$$V(t, X_t) = \max_{X_{t+1}} \{\mu(X_t - X_{t+1}) + \beta s_t V(t+1, X_{t+1})\} \quad (9)$$

$$\text{s.t. } X_t \geq X_{t+1} \geq 0 \quad (10)$$

5 Calibration

We assume the economy stays on a balanced growth path until 1993, the first year we have estimates of permanent and transitory income variances and income and consumption variances in the data. Starting from 1994, we assume the rest of the exogenous changes became effective as a surprise, but moving forward, households have perfect foresight of the changes in the future. The life-cycle model for each birth cohort is solved backwards starting from the last period of life using an endogenous grid point method proposed by [Carroll \(2006\)](#). Further details are provided in [Appendix C](#). We solve for rural and urban households separately and simulate the model for 10,000 households of each group. In 1993, we solve the decision rules for each cohort in 1993 in the transition and simulate forward using the updated decisions rules.

The model contains constant parameters and time-varying parameters that we calibrate in two stages. In the first stage, we choose some parameters from the literature or the data. In the second stage, we calibrate the remaining exogenous time-varying parameters to match certain moments along the transition path.

5.1 First-Stage Calibration

In the first stage, we select parameters from our data and the literature. Some parameters are fixed, while the rest are allowed to change over time. [Table 3](#) reports the first-stage calibrated parameters in the benchmark model.

5.1.1 Parameters fixed over time

Following the literature, we set the time discount factor β to 0.97. We set the coefficient of relative risk aversion (CRRA) γ to 3 for both urban and rural households, which is in the range $1 < \gamma < 6$. The annual real interest rate r is set to 1.4%, consistent with the nominal one-year bank deposit rate deflated by annual CPI inflation. Initial wealth is assumed to be 0. We let urban households start their life cycles with a level of permanent income that is 50% higher than rural households, consistent with evidence in our empirical data.

We set the retirement age R to be age 59 based on sharp declines in the labor participation rate after age 50 in both the urban and rural samples.⁹ The urban-rural age-specific survival probability s is borrowed from Zimmer et al. (2007).¹⁰ Figure B.2 in Appendix B.2 plots the age-dependent survival rate for urban and rural samples. Note that households are assumed to live with certainty until age 60 and begin facing a survival probability until the age of 100 when they die with certainty. We borrow the rural pension replacement rate from Dotsey et al. (forthcoming), and set it at 6 percent. Then, we take the replacement rates over the years 1993–2009 for urban workers from Fang and Feng (2018), and find the average pension replacement rate to be approximately 60 percent.

5.1.2 Parameters variant over time

In our model, we allow the rural and urban age-dependent income growth rate to change after 1997.¹¹ To characterize the life-cycle income process in our model, we estimate the baseline age-specific deterministic income growth rate from the CHNS data, separately for urban and rural households. The detailed estimation process can be found in Appendix B.3. To form the estimation sample, we keep households with heads between age 25 and 60. We then regress the log of household disposable income on a set of cohort dummies,

⁹We use participants' answers to the question "Are you presently working/employed?" in the CHNS survey questionnaire to define labor participation. The plots of labor participation against age can be found in Appendix B.1.

¹⁰We make use of their mortality rate derived from CHNS for data consistency. Their CHNS data covers six waves (1989, 1991, 1993, 1997, 2000, and 2004), and overlaps with most of the waves used in our dataset. Their data also separates urban and rural residences, allowing us to model urban and rural households separately. They document substantially higher mortality rates for the elderly in rural areas than urban areas. Specifically, the survival rate for agents aged 60–64 is 0.991 (0.985), 65–69 is 0.979 (0.973), 70–74 is 0.971 (0.951), and 75–79 is 0.949 (0.936) for the urban (rural) sample.

¹¹We use the year of 1997 as the cutoff as it sits around the middle of the years we have data for.

Table 3. Parameters Calibrated in First Stage

Parameter	Description	Rural	Value	Urban	Target (Source)
Fixed Parameters					
γ	coefficient of relative risk aversion		3		
β	time discount factor		0.97		Yang (2023)
r	annual real interest rate		0.014		Yang (2023)
W_0	initial permanent income	1.0		1.5	CHNS
a_0	initial wealth		0		Model assumption
R	retirement age		59		CHNS
s_t	age-dependent survival rate	Figure B.2 in Appendix B.2			Zimmer et al. (2007)
η	pension replacement ratio	0.06		0.6	Dotsey et al. (forthcoming) Fang and Feng(2018)
Time-Varying Parameters					
g_t	age-specific income growth rate before 1997	Figure in Appendix B.3			CHNS
Δg	increase in income growth rate after 1997	0.5%		1.0%	CHNS

household size, and a fourth-order polynomial of the age of the household head. The predicted marginal effect of age on household income at each age is the income growth rate at the corresponding age.

To compare the difference in the average income growth rates in the pre-reform era (prior to 1997) and after, we start with computing the average income by cohort and by year. Then, we compute the income growth rates separately for each cohort and for the periods of 1993–1996 and 1997–2009. The average income growth rate is the average across cohorts in each period. To facilitate comparison, we restrict household head age to 25–60 in the samples. Based on the calculation, the data reports average income growth rates of 0.031 (0.036) in 1993–1997 and 0.085 (0.052) in 1997–2006 for the urban (rural) sample. To capture the effect of a higher income growth rate on household saving behavior, we consider the difference in the average growth rates in the two periods as the rise in income growth rate in the model. To avoid exaggerating the effects, we implement one-fourth of the computed differences and assume this takes place in the year 1998. In effect, we assume the income growth rate to be 1.0 (0.5) percentage point higher for urban (rural) households at each age in the post-reform era than the baseline.

Figure B.3 in Appendix B.3 plots the gross income growth rate against household head age. The estimated annual income growth rate (by age) for urban households is on average about 2.2% for households aged between 25–29, 3.9% for those in their thirties, 3.9% for those in their forties, and 2.6% for those in their fifties. These numbers for rural households are 4.4%, 4.3%, 4.2%, and 1.5%, respectively.

5.2 Second-Stage Calibration

The second-stage calibration includes a set of parameters for the initial steady state and time-varying processes that we chose to match selected economic moments and time series along the transition path.

5.2.1 Parameters for the initial steady state

The remaining parameters to be determined are the benchmark permanent and transitory variances for the rural and urban samples. We calibrate our model to the steady state economy benchmark in 1993 to match the income variances in the 1993 data. In particular, given the empirical estimates of permanent and transitory income variances, we select the proportion of empirical variances to match our benchmark income inequalities, thus treating the remaining proportion as measurement error.

5.2.2 Parameters along the transition

In the following several years of which we have empirical data, we let the model determine a proportion of the empirical variances to best match income variance in the data, thus attributing the remaining portion of the empirical variances to measurement error. Consumption inequality is endogenously determined in the model. Table 4 reports the parameters estimated in the second stage.

Table 4. Parameters Calibrated in Second Stage

Parameter	Description	Year	Rural		Urban		Source
			Perm.	Tran.	Perm.	Tran.	
$\sigma_{\theta_t}^2, \sigma_{v_t}^2$	variances of income	1993	0.025	0.098	0.016	0.099	CHNS
		1994-7	0.022	0.149	0.030	0.085	
		1998-2000	0.036	0.251	0.046	0.325	
		2001-04	0.057	0.27	0.063	0.326	
		2005-06	0.054	0.150	0.057	0.094	

6 Initial Urban-Rural Differences

The initial steady state economies are calibrated separately for rural and urban households to match their respective data moments. In Table 5, we present data moments and their corresponding model moments. Our benchmark model does a reasonable job at matching the income and consumption variances in the base year. In particular, our

model captures the fact that in the base year, income inequality among rural households is substantially higher than that among their urban counterparts, whereas the consumption inequality in the rural sample is only slightly higher than that in the urban sample. Our model generates higher consumption variances than those in the data for both rural and urban households. Given our model specification that the only tool available for agents to insulate their consumption is personal buffer-stock savings, the discrepancy between the theoretical and empirical consumption inequalities can be understood by the existence of other insurance tools available to the actual economic agents, such as labor supply and retirement decisions, that are not present in the model (Jappelli and Pistaferri, 2014).

Table 5. Income and Consumption Inequality in 1993

	Income		Consumption	
	Data	Model	Data	Model
Urban	0.45	0.44	0.19	0.30
Rural	0.89	0.90	0.24	0.35

Notes: Model-implied variance of logarithms of disposable income and consumption in 1993.

To quantify how much each factor has contributed to the difference in income and consumption inequality in rural and urban samples, we carry out the following decomposition experiment. Starting from the baseline urban model, we sequentially replace each parameter that differs between the urban and rural calibrations with its rural counterpart, allowing their effects to accumulate until all parameters are switched.

Table 6. Decomposition: Initial Rural and Urban Income and Consumption Inequality, Saving Rate

	Income	Consumption	Saving rate
<i>Urban</i>	<i>0.44</i>	<i>0.30</i>	<i>0.17</i>
+ replace age-specific survival rate (s_t) ↓	0.44	0.30	0.16
+ replace income growth rate (g_t) ↑	0.46	0.31	0.17
+ replace pension replacement ratio (η) ↓	0.76	0.24	0.25
+ replace initial permanent income (W_0) ↓	0.76	0.24	0.25
+ replace transitory income variance	0.76	0.24	0.25
+ replace permanent income variance ↑	0.90	0.35	0.29
<i>Rural</i>	<i>0.90</i>	<i>0.35</i>	<i>0.29</i>

Notes: Numbers are model-implied variances of log income and consumption, and saving rate. To derive the results in the table above, we start from the model's benchmark results for urban sample and then replace the value of each parameter used in the urban sample to the one used in rural counterpart, if they are different between those two samples. Tables 3 and 4 contain the parameter values employed in the rural and urban samples, respectively.

We present the results in Table 6 and highlight the main contributors to urban-rural differences in income inequality and consumption insurance. Starting from the benchmark

results for the urban sample, we first consider the impact of different urban-rural age-specific survival rate. Rural households have a lower survival rates than their urban counterparts at all ages older than 60 (see Appendix B.2). When we decrease the urban survival rate to the rural level, the impact on income variance, consumption variance, and savings rate is negligible.

We then change the age-specific income growth rate for the urban sample to that of the rural sample (shown in Figure B.3 in Appendix B.3). Changing the life-cycle income growth rate for urban households increases income variance from 0.44 to 0.46, while consumption variance and saving rate slightly increase. To understand the results, in Figure 2, we plot the life-cycle profiles of log income and consumption before and after the shift of income growth rate. Due to the higher rural income growth rate and the tendency for consumption to track income along the life cycle, we observe a higher cumulative growth rate of consumption along the life cycle. In other words, there are larger variations in income and consumption by age in rural relative to urban areas. Thus, using urban income growth would generate lower income variance and consumption variances.

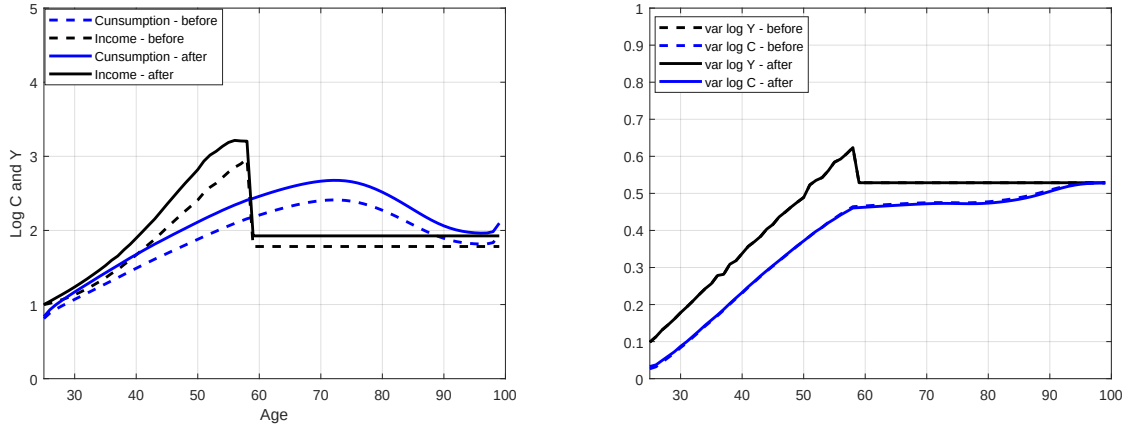


Figure 2. Model Simulated Life-cycle Profile of Log Income and Consumption Prior to vs. after Income Growth Rate Shift

Notes: The left panel shows life-cycle profiles of levels of log income and consumption, the right panel shows variances log income and consumption by age. The dashed lines represent life cycle profiles prior to the income growth rate shift, the solid lines represent life cycle profiles after the shift, respectively.

Replacing the pension replacement ratio from 0.6 to 0.06 drastically changes the variances in income and consumption. Reducing the urban pension replacement ratio from 0.60 to 0.06 increases the overall income variance from 0.46 to 0.76, but the overall consumption variance decreases from 0.31 to 0.24. The average savings rate, on the other

hand, increases from 0.17 to 0.25. As shown in Figure 3, there are two driving forces behind the decrease in consumption variance. First, the left panel shows as pension replacement ratio decreases, urban households decrease consumption, especially those in the middle or old ages, young households also reduce their consumption, but to a lesser extent. As a result, we observe a flatter life-cycle consumption profile after pension reduction relative to before, leading to smaller consumption variations between age groups.

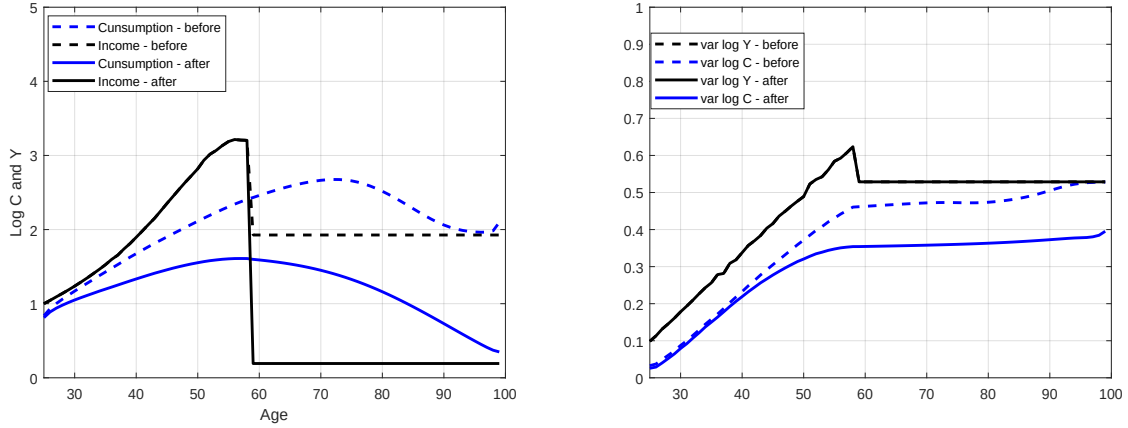


Figure 3. Model Simulated Life-cycle Profile of Log Income and Consumption Prior to vs. after Pension shift

Notes: The left panel shows life-cycle profiles of levels of log income and consumption, the right panel shows variances log income and consumption by age. The dashed lines represent life cycle profiles prior to the pension replacement ratio shift, the solid lines represent life cycle profiles after the shift, respectively.

Lower initial permanent income (decreased from 1.5 to 1) has no obvious impact on model output. Since transitory income variances are close in both samples, it has no clear impact on the urban-rural variance gaps.

Permanent income variance, on the other hand, is much higher for rural households (0.025) than urban households (0.016). When permanent income variance is increased from 0.016 to 0.025, the overall income variance increases from 0.76 to 0.90 and consumption variance increases from 0.24 to 0.35. The average saving rate increases by 4 percentage points from 25 percentage to 29 percent. As shown in Figure 4, as future income becomes more uncertain, younger households who started out with zero asset need to reduce consumption and increase saving to build up a stock of asset to buffer potential future income losses. Older households, on the other hand, can enjoy higher consumptions because they already built up their assets. These dynamics create greater consumption variations between age groups, leading to higher overall consumption inequality. The increase in the savings rate among the young drives increases in the average savings rate.

To this point, we have found that differences between urban and rural households in the pension replacement ratio and in the magnitude of permanent income shocks are the most prominent contributors to urban–rural inequality. While the effects of these parameter changes may vary slightly depending on the order in which they are introduced, due to nonlinearities, this exercise nevertheless highlights the outsized impact of the pension replacement ratio and permanent income shocks.

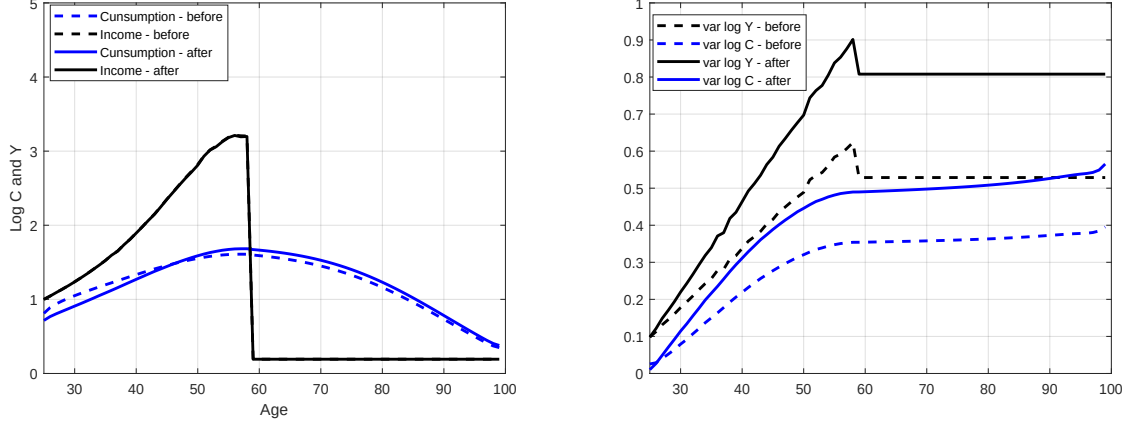


Figure 4. Model Simulated Life-cycle Profile of Log Income and Consumption Prior to vs. after Permanent Variance Shift

Notes: The left panel shows life-cycle profiles of levels of log income and consumption, the right panel shows variances log income and consumption by age. The dashed lines represent life cycle profiles prior to the permanent income variance shift, the solid lines represent life cycle profiles after the permanent variance shift, respectively

7 Urban-Rural Differences Over Time

The previous section investigates how rural and urban households differ initially in income and consumption inequality. After 1993, our benchmark model considers evolving permanent and transitory income variances, rising income growth rates, and their impact on income and consumption inequalities. We first present how these factors influence the evolution of income and consumption inequality over time, and then examine their impact on saving rates.

7.1 Consumption Inequality Over Time

Figure 5 displays the model-generated cross-sectional income and consumption inequalities for the rural and urban samples. The model successfully reproduces two salient facts

in the data. First, consumption inequality is always below income inequality (for both samples). Second, both income inequality and consumption inequality increase over time (for both samples), but the increase in consumption inequality is much flatter than in income inequality. The model-implied increase in income inequality is in line with the empirical data, from 0.90 (0.44) in 1993 to 1.22 (0.83) in 2009 for rural (urban) households, compared to 0.89 (0.45) to 1.25 (0.85) in the data.

Consumption inequality increased due to income shocks not being fully insurable. The growth in consumption inequality was less pronounced than in income inequality because households used personal savings to partially insure their consumption. Similar findings can be seen in [Storesletten et al. \(2004\)](#) and [Krueger and Perri \(2006\)](#) for the U.S. economy. Our model-implied consumption inequality is 0.38 (0.30) in 1993 and 0.50 (0.54) in 2009 for rural (urban) households, a 0.12 (0.24) log point theoretical increase, compared to the 0.20 (0.12) log point increase in the data. Our model does a reasonable job of matching the increase in income inequality. However, the model slightly underpredicts the increase in consumption inequality for the rural sample and overpredicts the increase in consumption inequality for the urban sample.

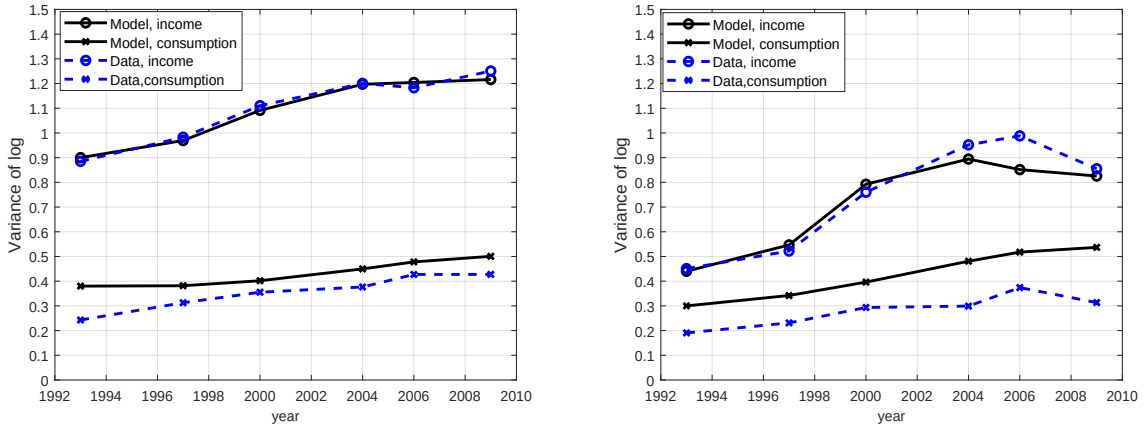


Figure 5. Income and Consumption Inequality (Variance of Logs), CHNS vs. Model

Notes: The graph plots the model-implied income and consumption inequality in the cross section against the data for rural (left) and urban (right) households. The blue dashed lines represent the empirical cross-sectional income inequality and consumption inequality. The black lines represent the theoretical cross-sectional income inequality and consumption inequality.

A detailed comparison of the income and consumption Gini coefficients between the empirical data and theoretical model can be found in Figure 6. The model slightly overpredicts the levels of the income and consumption Gini coefficients for both samples. The increase in the income Gini coefficients for rural (urban) households from 1993 to

2009 is 0.10 (0.12) in the model and 0.08 (0.05) in the data, while the consumption Gini coefficients increased by 0.06 (0.11) in the model and 0.06 (0.03) in the data.

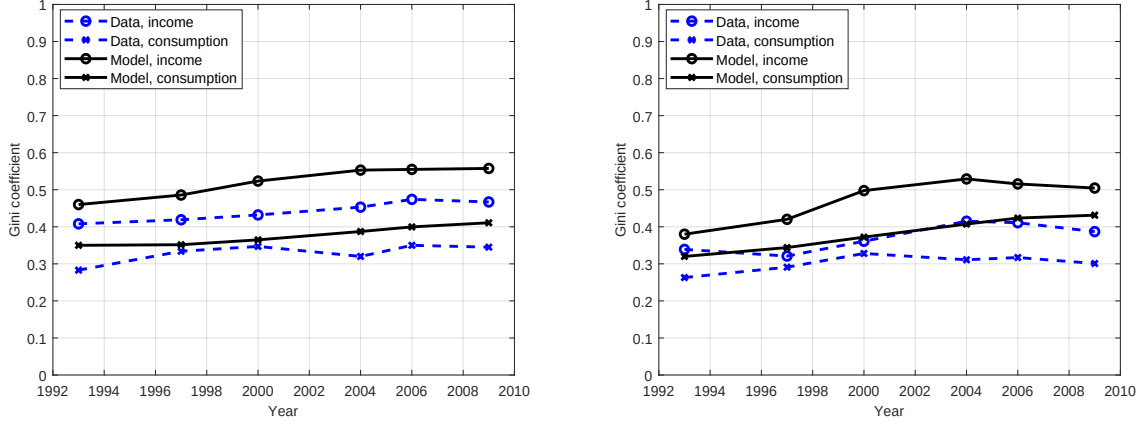


Figure 6. Income and Consumption Gini Coefficient, CHNS vs. Model

Notes: The graph plots the model-implied income and consumption Gini coefficient in the cross section against the data for rural (left) and urban (right) households. The blue dashed lines represent the empirical cross-sectional income and consumption Gini. The black lines represent the theoretical cross-sectional income and consumption Gini.

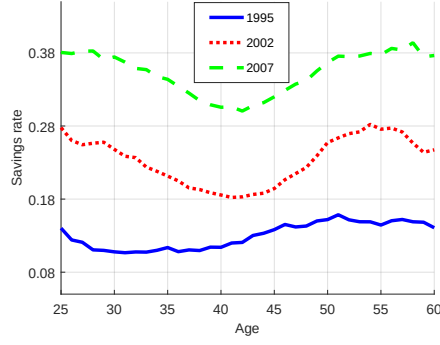
7.2 Aggregate Saving Rates Over Time

Table 7 shows the empirical and model-implied average saving rate by year for urban and rural households. The saving rate in the empirical data, computed from the Chinese Household Income Projects dataset, increased substantially for urban households but decreased for rural households.¹² Our benchmark model generates an increasing average saving rate over time for both samples. Compared with the empirical data, the model generates a higher average saving rate in the cross section.

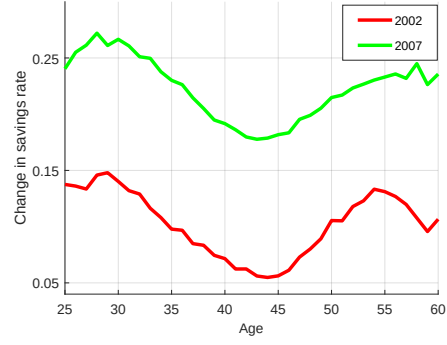
Table 7. Saving Rate: Model vs. Data				
	Urban		Rural	
Year	Data	Model	Data	Model
1995	0.12	0.17	0.27	0.29
2002	0.22	0.33	0.18	0.28
2007	0.35	0.53	0.17	0.43

Notes: Saving rate in the data is calculated from Chinese Household Income Projects. Details of the calculation can be found in Appendix D.

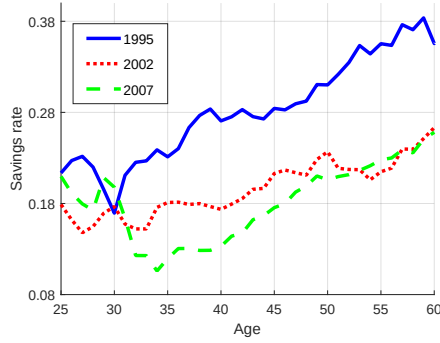
¹²Saving rate at each age is calculated as $1 - (Consumption)/(Income)$ where income and consumption are the averages of income and consumption at each age. See Appendix D for details.



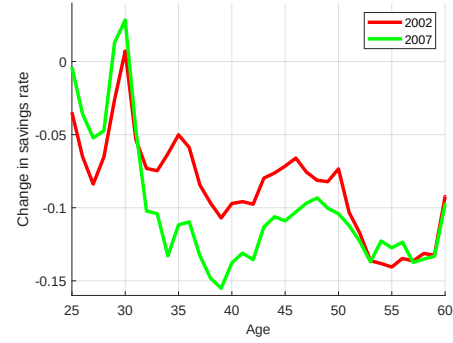
(a) Age-saving rate profile, Urban



(b) Change in saving rates, Urban



(c) Age-saving rate profile, Rural



(d) Change in saving rates, Rural

Figure 7. Cross-Sectional Average Saving Rates by Age of the Household Head

Notes: Panels (a) and (b) correspond to urban households, and panels (c) and (d) correspond to rural households. The age profile of household saving rate was smoothed by applying a 9-year age bin, that is, the average saving rate at each age is calculated by combining households with those 4 years younger and older. Saving rate is equal to $1 - (Consumption)/(Income)$. Changes in saving rate are relative to the 1995 value. Data is from Chinese Household Income Projects (1995, 2002, 2007).

We then show the average household saving rate against age in the cross section. Panels (a) and (b) in Figure 7 plot the cross-sectional average saving rates by age of the household head and the change in saving rates for the urban sample. Two patterns stand out. First, the age-saving rate profile started relatively flat in 1995 and became increasingly U-shaped over time as young and old households increased their saving rate more than the middle-aged. Second, there is an across-the-board increase in saving rates, with the saving rate increasing at every age. Specifically, urban households saved at a rate at least 5 and 17 percentage points higher in 2002 and 2007, respectively, than in 1995.

Panels (c) and (d) in Figure 7 plot the age-saving rate profile and the change in saving rates in the cross section among rural households. In terms of the rural age-saving rate profile in 1995, the saving rate first slightly decreases from age 25 to 30 before it rises continuously through the age of 60. Compared to 1995, those aged between 35 and 60

saved at a rate at least 5 and 9 percentage points lower in 2002 and 2007, respectively.

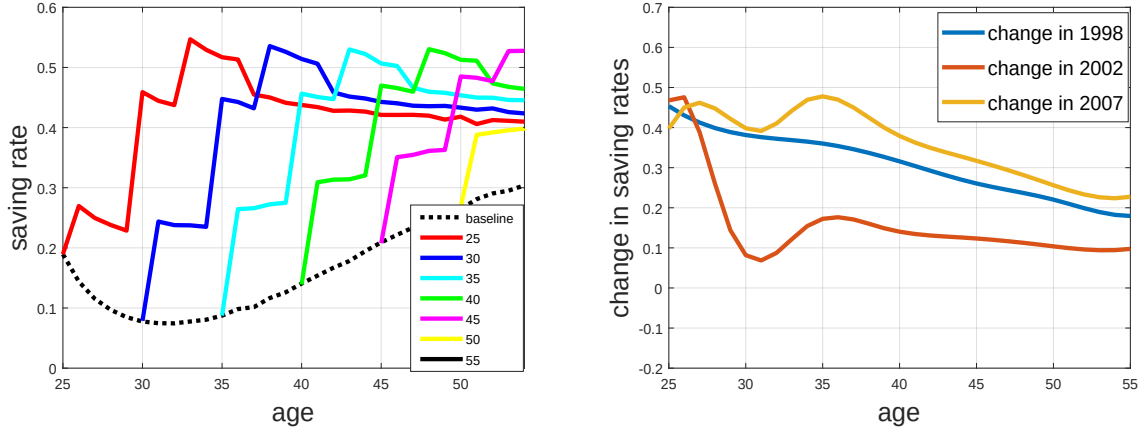


Figure 8. Average Saving Rates by Age, Urban

Notes: The dotted line in the left panel corresponds to the cross-sectional saving rate in 1993. Each other line indicates the life-cycle (going beyond 1993) saving rate for households of different 1993 ages. In the right panel, each line captures the change in saving rates from 1993 for each birth cohort indicated by age.

We now show the average household saving rate against age from our benchmark model. Figure 8 shows the model simulation results for urban households. The dotted line in the left panel corresponds to the saving rates in 1993 where age corresponds to households' age in 1993. This can be compared with the cross-sectional profile in panel (a) of Figure 7. The baseline for the model is given by treating 1993 as the steady state, so the dotted line is also the baseline life-cycle saving rates for each birth cohort alive in 1993. After shocks are introduced, households re-optimize their life-cycle saving rates accordingly by changing their consumption and saving decisions. Each colored line in the left panel of Figure 8 indicates the shock-endogenized life-cycle saving rate for households of different 1993 ages. For example, the red line shows the saving rate of urban households who were age 25 in 1993 as they age, distinct from aging in the steady state (the dotted line).

As there is an increase in permanent income variance (despite a decrease in transitory income variance) in 1994, in the model, households respond by immediately increasing saving rates from the baseline. As both permanent and transitory income variances drastically increased in 1998, all households responded by increasing their saving rate in the fifth year of the life cycle, with the increase decreasing with age. Since young households have not built a large stock of wealth to buffer against potential future income losses, and the old are nearing retirement, both groups save more once income becomes

more uncertain. The increase in permanent income variances continued through 2001 (before it slightly dropped in 2005), inducing further increases in the saving rates among urban households.

The model assumes shocks to income variances start to take place in 1994, but the one-time shock to income growth rate takes place in 1997. The implied cross-sectional changes in saving rates, after households have adjusted for shocks, are shown in the right panel of Figure 8. From the base-year saving rates in 1993, we derive changes in saving rates in the cross section for years 1998, 2002, and 2007 in which households have aged five, nine, and fourteen years, respectively, in their life cycle. Compared to the saving rate in the 1993 base year, the implied cross-sectional average change in saving rates among urban households is 16 and 20 percentage points in 2002 and 2007, respectively.

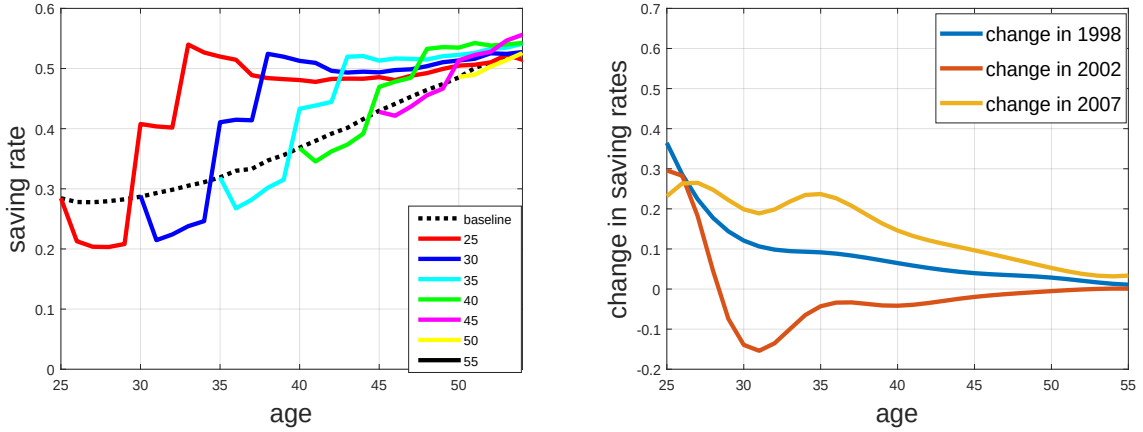


Figure 9. Average Saving Rates by Age, Rural

Notes: The dotted line in the left panel corresponds to the cross-sectional saving rate in 1993. Each other line indicates the life-cycle (going beyond 1993) saving rate for households of different 1993 ages. In the right panel, each line captures the change in saving rates from 1993 for each birth cohort indicated by age.

Figure 9 presents the model results from analogous experiments with rural households when all income shocks are introduced into the model simultaneously (including evolving income variances and a one-off 0.5% shift in income growth rate after 1997). Due to the slight decrease in theoretical permanent income variance in 1994 (though transitory income variance increased, its positive effect on saving rate is completely offset by the negative pressure placed by permanent income variance), rural households experienced an immediate drop in saving rates. On the other hand, since both permanent and transitory income variances nearly doubled in 1998 relative to 1994, we witness drastic increases in saving rate across all ages in the fifth year of the life cycle, with the largest increases

observed among young households. While the increase in saving rate from 1995 (the baseline) to 1998 is approximately 8 percentage points for the average household, it is only 2 percentage points for the average 50-year-old. Since permanent income variance continued to increase in 2001 before it slightly dropped in 2005, saving rates further increased in the eighth year of the life cycle but eventually dropped in the twelfth year of the life cycle across households.

The right panel of Figure 9 captures the change in saving rate in the cross section. Relative to the saving rate in the 1995 base year, the implied cross-sectional average change in saving rates among rural households is -1 and 14 percentage points in 2002 and 2007, respectively.

7.3 Decomposition

In our benchmark model, we jointly capture the effects of 1. changes to permanent and transitory income variances and 2. rising income growth rates on inequality metrics and savings rates. To reveal the contribution of each change to the patterns observed in income inequality and consumption inequality, we introduce one change at a time while holding all other parameters at their baseline values in 1993.

7.3.1 Permanent income risks

We start with only allowing the permanent income risks to vary while holding all other factors at their baseline levels. Figure 10 depicts the model-implied inequality, measured in log variance. The increase in permanent income variance can largely generate all the observed patterns in Figure 5. In the model, the rising permanent income variance in the rural sample alone implies a 0.22 point increase (from 0.90 to 1.12) in income inequality from 1993 to 2009, compared to a 0.31 point increase from 0.90 to 1.21 in the benchmark model. The rising permanent income variance in the urban sample alone implies a 0.40 point increase (from around 0.44 to 0.83) in income inequality, closely following the benchmark model increase from 0.44 to 0.83. The model-implied increase in consumption inequality among rural (urban) households is around 0.10 (0.21) points from 0.38 (0.30) to 0.48 (0.51), close to that in the benchmark model from 0.38 (0.30) to 0.50 (0.54). These observations led us to believe that the rise in permanent income variance alone can explain almost all of the rise in income inequality. Second, the fact that

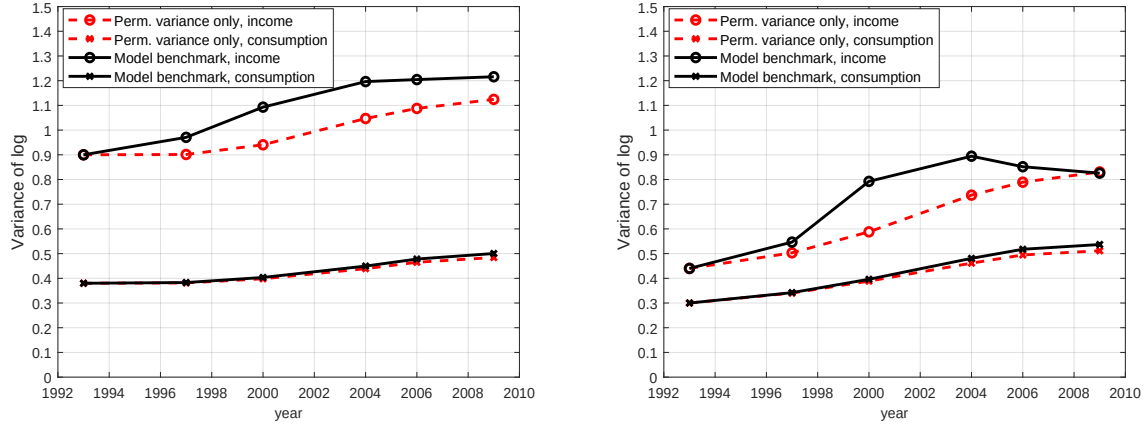


Figure 10. Income and Consumption Inequality due to Permanent Income Risks

Notes: This graph plots the model-implied income and consumption inequality in the cross section against the benchmark model for rural (left) and urban (right) households. The black lines represent the benchmark theoretical cross-sectional income inequality and consumption inequality which reflects the combined effects of all the income shocks considered. The red dashed lines represent the theoretical cross-sectional income inequality and consumption inequality when there is variation in permanent income variance only.

consumption inequality also increases as the result of the increasing permanent income variations shows permanent variance is only partially insured.

Table 8 shows the income and consumption Gini coefficients by year in the benchmark model and in each decomposition. The second panel shows that permanent income shocks are largely passed through to consumption. Between 1993 and 2009, the change in permanent variance alone implies a 0.07 (0.13) point increase in the income Gini coefficients from 0.46 (0.38) to 0.53 (0.51) for the rural (urban) sample. The consumption Gini coefficients increased by 0.06 (0.10) points from 0.34 (0.32) to 0.40 (0.42) for rural (urban) households.

Table 9 displays the saving rate by year in the benchmark model and in each decomposition. The second panel shows that rising permanent income variance induced a 14 (27) percentage point increase from 1995 to 2007 in average saving rate for rural (urban) households.

7.3.2 Transitory income risks

Figure 11 shows the result of only allowing transitory income variance to change while all other parameters remain at baseline assumptions. Due to the increase in transitory variance before 2000, our model implies a 0.15 (0.21) point increase in income variance for rural (urban) households from 0.90 (0.44) to 1.05 (0.65) between 1993 and 2000.

Table 8. Model-Implied Income and Consumption Gini

	1993	1997	2000	2004	2006	2009
<i>Panel I. Model benchmark</i>						
<i>Rural</i>						
Income Gini coefficients	0.46	0.49	0.52	0.55	0.55	0.56
Consumption Gini coefficients	0.35	0.35	0.36	0.39	0.40	0.41
<i>Urban</i>						
Income Gini coefficients	0.38	0.42	0.50	0.53	0.52	0.50
Consumption Gini coefficients	0.32	0.34	0.37	0.41	0.42	0.43
<i>Panel II. Variants of permanent income variance only</i>						
<i>Rural</i>						
Income Gini coefficients	0.46	0.46	0.47	0.51	0.52	0.53
Consumption Gini coefficients	0.34	0.34	0.35	0.37	0.38	0.40
<i>Urban</i>						
Income Gini coefficients	0.38	0.41	0.44	0.48	0.50	0.51
Consumption Gini coefficients	0.32	0.34	0.37	0.40	0.42	0.42
<i>Panel III. Variants of transitory income variance only</i>						
<i>Rural</i>						
Income Gini coefficients	0.46	0.49	0.51	0.51	0.50	0.49
Consumption Gini coefficients	0.34	0.34	0.34	0.35	0.34	0.34
<i>Urban</i>						
Income Gini coefficients	0.38	0.40	0.45	0.48	0.41	0.38
Consumption Gini coefficients	0.32	0.32	0.33	0.33	0.32	0.32
<i>Panel IV. Variants of income growth rate only</i>						
<i>Rural</i>						
Income Gini coefficients	0.46	0.46	0.46	0.46	0.46	0.46
Consumption Gini coefficients	0.34	0.34	0.34	0.34	0.34	0.34
<i>Urban</i>						
Income Gini coefficients	0.38	0.38	0.38	0.38	0.38	0.38
Consumption Gini coefficients	0.32	0.32	0.33	0.33	0.33	0.32

Notes: The first panel shows model results for the benchmark model which reflects the combined effects of all income shocks. The second panel shows model results when only permanent income variance is allowed to vary over time while holding all other parameters at baseline levels. The third panel shows model results when only transitory income variance is allowed to vary over time while holding all other parameters at baseline levels. The fourth panel shows model results when only the income growth rate is allowed to vary over time while holding all other parameters at baseline levels.

Due to the decrease in transitory variance in the post-2000 era in the model, income inequality decreases after 2000 in both samples. Consumption inequality, on the other hand, remained nearly constant throughout the period in both samples. These simulation results suggest that households are able to achieve full insurance against transitory variance through personal saving and self-insurance. [Blundell et al. \(2008\)](#) came to the same conclusion for households in the U.S. using data from the Panel Study of Income Dynamics.

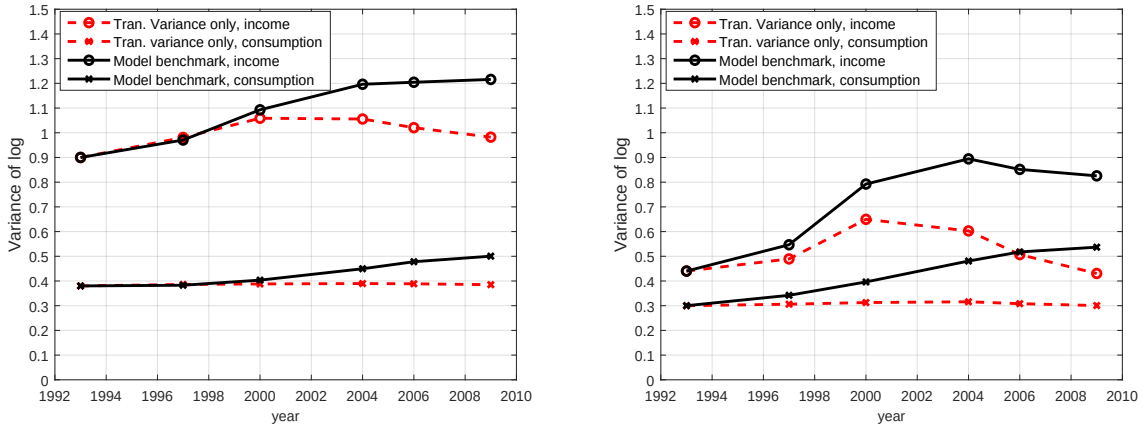
The yearly income and consumption Gini coefficients when only transitory variance

Table 9. Model-Implied Saving Rate

	1995	2002	2007
<i>Panel I. Model benchmark</i>			
Rural	0.29	0.28	0.43
Urban	0.23	0.30	0.53
<i>Panel II. Variants of permanent income variance only</i>			
Rural	0.29	0.29	0.43
Urban	0.17	0.38	0.54
<i>Panel III. Variants of transitory income variance only</i>			
Rural	0.29	0.29	0.30
Urban	0.17	0.17	0.21
<i>Panel IV. Variants of income growth rate only</i>			
Rural	0.29	0.27	0.27
Urban	0.17	0.13	0.13

Notes: The first panel shows model results for the benchmark model which reflects the combined effects of all income shocks. The second panel shows model results when only permanent income variance is allowed to vary over time while holding all other parameters at baseline levels. The third panel shows model results when only transitory income variance is allowed to vary over time while holding all other parameters at baseline levels. The fourth panel shows model results when only the income growth rate is allowed to vary over time while holding all other parameters at baseline levels.

varied can be found in the third panel of Table 8. The income Gini coefficients due to the evolution of transitory variance, similar to the behavior of income variance, first

**Figure 11.** Income and Consumption Inequality due to Transitory Income Risks

Notes: The graph plots the model-implied income and consumption inequality in the cross section against the benchmark model for rural (left) and urban (right) households. The black lines represent the benchmark theoretical cross-sectional income inequality and consumption inequality which reflects the combined effects of all the income shocks considered. The red dashed lines represent the theoretical cross-sectional income inequality and consumption inequality when there is variation only in transitory income variance.

increased from 1993 to 2000 but leveled off afterwards. This is true in both rural and urban samples. The consumption Gini coefficients remained nearly unchanged between

1993 and 2009 for both samples.

The third panel of Table 9 shows that in response to changes in transitory variance, saving rates rose by 3 percentage points for the urban sample and 1 percentage point for the rural sample between 1993 and 2009.

7.3.3 Income growth rate

In this section, we allow the income growth rate to shift in 1998 while fixing all other parameters at the baseline levels. That is, for the rural (urban) sample, the income growth rate is 0.5% (1%) higher than the baseline after 1998. As shown in Figure 12, when all other factors are held constant, the rising income growth rate alone shows little impact on either income inequality or consumption inequality. The income and consumption Gini coefficients (shown in the fourth panel of Table 8) also remained at their baseline levels in 1993 despite the increase in income growth rate after 1998.

The average saving rate, on the other hand, decreased by 4 (2) percentage points in the urban (rural) sample from 17 (29) percentage points to 13 (27) percentage points (shown in the fourth panel of Tables 9). In theory, the effect of a higher income growth rate on the saving rate is twofold, one direct, one indirect. On the one hand, at a given target wealth-to-income ratio (and thereby saving rate), a higher income growth rate directly contributes to an increase in the saving rate. On the other hand, all else held constant, a higher income growth rate tends to reduce households' target wealth-to-income ratio as they become less vulnerable to income risk. Therefore, the saving rate increases with the income growth rate if the direct effect dominates; otherwise, if the indirect effect dominates, a higher income growth rate will reduce the saving rate. The relative strength of the two effects cannot be determined analytically but must be assessed numerically by solving the model.¹³

To this end, we conclude that the increase in permanent income shocks adversely impacted income equality and the other main welfare measure, consumption equality, among Chinese households. The rise in permanent income variance has contributed to the increased income inequality, eroding households' ability to smooth their consumption. Our model results suggest that Chinese households only achieved partial insurance against permanent income variations but full insurance against transitory income variances. The

¹³See Yang (2023) for detailed exposition.

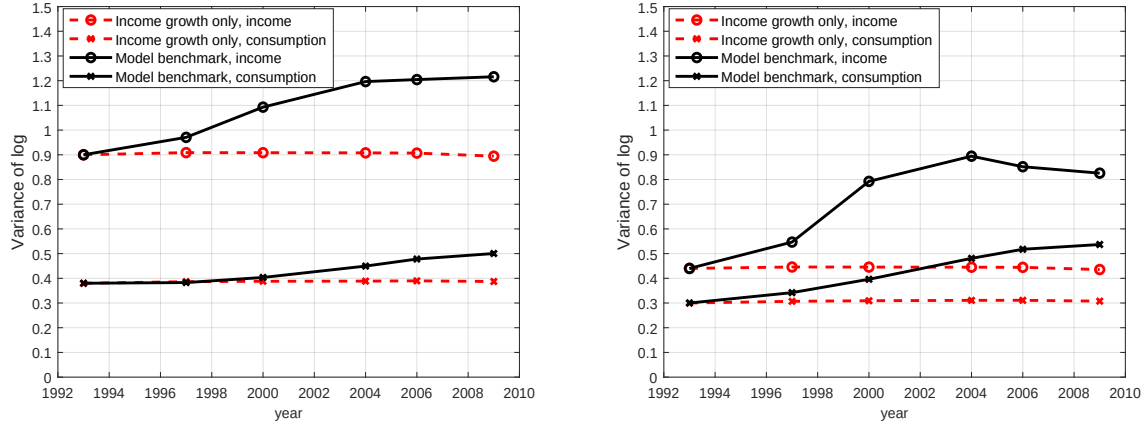


Figure 12. Income and Consumption Inequality due to Income Growth Changes

Notes: The graph plots the model-implied income and consumption inequality in the cross section against the data for rural (left) and urban (right) households. The black lines represent the benchmark theoretical cross-sectional income inequality and consumption inequality which reflects the combined effects of all the income shocks considered. The red dashed lines represent the theoretical cross-sectional income inequality and consumption inequality when there is variation only in the income growth rate.

accelerated income growth rate in the late 1990s and early 2000s led to a lower saving rate among households but made little contribution to income inequality and consumption inequality.

7.3.4 Pension Replacement Rate

This section presents the pension replacement rate shock on income inequality measures, for urban and rural households separately.

8 Sensitivity Analysis

In this section, we investigate how the coefficient of relative risk aversion affects consumption inequality and the saving rate in 1993 while keeping other parameters consistent with the benchmark.

The results in Table 10 show that the saving rate is the only metric that is sensitive to the level of the coefficient of risk aversion. The higher the coefficient is, the more risk-averse households are and the more they will save under the same level of income uncertainty. Consumption inequality, measured in both variance and Gini coefficient, reduces as a result of higher savings, but the change is very small. This indicates that our benchmark results on consumption inequality, measured both in variance of log and

Table 10. Model Variables by Coefficient of Relative Risk Aversion

γ	Income Variance	Consumption Variance	Income Gini coefficients	Consumption Gini coefficients	Saving Rate
<i>Urban</i>					
2	0.44	0.32	0.38	0.33	0.11
3 (baseline)	0.44	0.30	0.38	0.32	0.17
4	0.44	0.30	0.38	0.32	0.24
<i>Rural</i>					
2	0.90	0.36	0.46	0.34	0.20
3 (baseline)	0.90	0.35	0.46	0.34	0.29
4	0.90	0.35	0.46	0.35	0.34

Notes: Model results using baseline parameters in 1993 while allowing coefficient of risk aversion to vary from 2 to 4.

Gini coefficient, are not sensitive to the CRRA coefficient.

9 Conclusion

This paper documents a pronounced rise in both income and consumption inequality in China between 1989 and 2009, with income inequality increasing more sharply than consumption inequality. In this paper, we investigated how Chinese households are able to secure their consumption against increasingly uncertain income via personal saving as a self-insurance mechanism. We find that the main contributor to the rising inequality of income and consumption is the overall rising permanent income variance. We present evidence of partial insurance against permanent income variance and full insurance against transitory income variance among Chinese households.

In this paper, we focus on self-insurance through savings as the only consumption-smoothing mechanism. Other consumption-smoothing mechanisms may be available in the data, such as receiving monetary help from an extended family network, flexible labor supply decisions, amongst others. Regardless, we view our quantitative results as the upper bound of consumption smoothing as the presence of other consumption-smoothing mechanisms would further reduce consumption inequality.

References

- Mark Aguiar and Mark Bilis. Has consumption inequality mirrored income inequality? *American Economic Review*, 105(9):2725–56, 2015.
- Orazio P Attanasio and Guglielmo Weber. Consumption and saving: models of intertemporal allocation and their implications for public policy. *Journal of Economic literature*, 48(3):693–751, 2010.
- Abhijit Banerjee, Xin Meng, Tommaso Porzio, and Nancy Qian. Aggregate fertility and household savings in china. NBER Working paper 20050 Revised, 2023.
- Garry F Barrett, Thomas F Crossley, and Christopher Worswick. Consumption and income inequality in australia. *Economic Record*, 76(233):116–138, 2000.
- Richard Blundell and Ian Preston. Consumption inequality and income uncertainty. *The Quarterly Journal of Economics*, 113(2):603–640, 1998.
- Richard Blundell, Luigi Pistaferri, and Ian Preston. Consumption inequality and partial insurance. *American Economic Review*, 98(5):1887–1921, 2008.
- Loren Brandt and Thomas G Rawski. *China’s great economic transformation*. Cambridge university press, 2008.
- Matthew Brzozowski, Martin Gervais, Paul Klein, and Michio Suzuki. Consumption, income, and wealth inequality in canada. *Review of economic dynamics*, 13(1):52–75, 2010.
- Hongbin Cai, Yuyu Chen, and Li-An Zhou. Income and consumption inequality in urban china: 1992–2003. *Economic Development and Cultural Change*, 58(3):385–413, 2010.
- Christopher D Carroll. Buffer-stock saving and the life cycle/permanent income hypothesis. *The Quarterly journal of economics*, 112(1):1–55, 1997.
- Christopher D Carroll. The method of endogenous gridpoints for solving dynamic stochastic optimization problems. *Economics letters*, 91(3):312–320, 2006.
- Marcos Chamon and Eswar Prasad. Why are saving rates of urban households in china rising? *American Economic Journal: Macroeconomics*, 2(1):93–130, 2010.

- Marcos Chamon, Kai Liu, and Eswar Prasad. Income uncertainty and household savings in china. *Journal of Development Economics*, 105:164–177, 2013.
- Horag Choi, Steven Lugauer, and Nelson C Mark. Precautionary saving of chinese and us households. *Journal of Money, Credit and Banking*, 49(4):635–661, 2017.
- Taha Choukhmane, Nicolas Coeurdacier, and Keyu Jin. The one-child policy and household saving. *Journal of the European Economic Association*, 21(3):987–1032, 2023.
- Chadwick C Curtis, Steven Lugauer, and Nelson C Mark. Demographic patterns and household saving in china. *American Economic Journal: Macroeconomics*, 7(2):58–94, 2015.
- Haiyan Ding and Hui He. A tale of transition: An empirical analysis of economic inequality in urban china, 1986–2009. *Review of Economic Dynamics*, 29:106–137, 2018.
- Michael Dotsey, Wenli Li, and Yang. Demographic transition, industrial policies, and chinese economic growth. *Quantitative Economics*, forthcoming.
- Hanming Fang and Jin Feng. The chinese pension system. Technical report, National Bureau of Economic Research Cambridge, MA, USA:, 2018.
- Giulio Fella, Serafin Frache, and Winfried Koeniger. Buffer-stock saving and households’response to income shocks. *International Economic Review*, 2016.
- Jonathan D Fisher, David S Johnson, and Timothy M Smeeding. Measuring the trends in inequality of individuals and families: Income and consumption. *American Economic Review*, 103(3):184–88, 2013.
- Suqin Ge, Yang, and Zhang. Population policies, demographic structural changes, and the chinese household saving puzzle. *European Economic Review*, 101:181–209, 2018.
- Pierre-Olivier Gourinchas and Jonathan A Parker. Consumption over the life cycle. *Econometrica*, 70(1):47–89, 2002.
- Hui He, Feng Huang, Zheng Liu, and Dongming Zhu. Breaking the ‘iron rice bowl’: Evidence of precautionary savings from the chinese state-owned enterprises reform. *Journal of Monetary Economics*, 94:94–113, 2018.

- Jonathan Heathcote, Fabrizio Perri, and Giovanni L Violante. Unequal we stand: An empirical analysis of economic inequality in the united states, 1967–2006. *Review of Economic dynamics*, 13(1):15–51, 2010.
- Jonathan Heathcote, Kjetil Storesletten, and Giovanni L Violante. Consumption and labor supply with partial insurance: An analytical framework. *American Economic Review*, 104(7):2075–2126, 2014.
- Jonathan Heathcote, Fabrizio Perri, Giovanni L Violante, and Lichen Zhang. More unequal we stand? inequality dynamics in the united states, 1967–2021. *Review of Economic Dynamics*, 50:235–266, 2023.
- Ayşe Imrohoroglu and Kai Zhao. The chinese saving rate: Long-term care risks, family insurance, and demographics. *Journal of Monetary Economics*, 96:33–52, 2018.
- Tullio Jappelli and Luigi Pistaferri. Does consumption inequality track income inequality in italy? *Review of Economic Dynamics*, 13(1):133–153, 2010.
- Tullio Jappelli and Luigi Pistaferri. Fiscal policy and mpc heterogeneity. *American Economic Journal: Macroeconomics*, 6(4):107–36, 2014.
- Greg Kaplan and Giovanni L Violante. How much consumption insurance beyond self-insurance? *American Economic Journal: Macroeconomics*, 2(4):53–87, 2010.
- Dirk Krueger and Fabrizio Perri. Does income inequality lead to consumption inequality? evidence and theory. *The Review of Economic Studies*, 73(1):163–193, 2006.
- Shi Li, Hiroshi Sato, and Terry Sicular. *Rising inequality in China: Challenges to a harmonious society*. Cambridge University Press, 2013.
- Josep Pijoan-Mas. Precautionary savings or working longer hours? *Review of Economic dynamics*, 9(2):326–352, 2006.
- Giorgio E Primiceri and Thijs Van Rens. Heterogeneous life-cycle profiles, income risk and consumption inequality. *Journal of monetary Economics*, 56(1):20–39, 2009.
- Raul Santaella-Llopis and Yu Zheng. The price of growth: consumption insurance in china 1989–2009. *American Economic Journal: Macroeconomics*, 10(4):1–35, 2018.

- Zheng Song, Kjetil Storesletten, and Fabrizio Zilibotti. Growing like china. *American economic review*, 101(1):196–233, 2011.
- Kjetil Storesletten, Christopher I Telmer, and Amir Yaron. Consumption and risk sharing over the life cycle. *Journal of monetary Economics*, 51(3):609–633, 2004.
- Shang-Jin Wei and Zhang. The competitive saving motive: Evidence from rising sex ratios and savings rates in china. *Journal of political Economy*, 119(3):511–564, 2011.
- Yu Xie and Xiang Zhou. Income inequality in today’s china. *Proceedings of the National Academy of Sciences*, 111(19):6928–6933, 2014.
- Dan Yang. Income growth, income uncertainty, and urban–rural household savings rate in china. *Regional Science and Urban Economics*, 99:103855, 2023.
- Dennis Tao Yang. Urban-biased policies and rising income inequality in china. *American Economic Review*, 89(2):306–310, 1999.
- Longmei Zhang, Ray Brooks, Ding Ding, Haiyan Ding, Jing Lu, and Rui Mano. *China’s High Savings: Drivers, Prospects, and Policies*. International Monetary Fund, 2018.
- Xiaodong Zhu. Understanding china’s growth: Past, present, and future. *Journal of Economic Perspectives*, 26(4):103–24, 2012.
- Zachary Zimmer, Toshiko Kaneda, and Laura Spess. An examination of urban versus rural mortality in china using community and individual data. *The Journals of Gerontology Series B: Psychological Sciences and Social Sciences*, 62(5):S349–S357, 2007.

Appendix

A Empirical Results

In this appendix, we provide additional materials related to Section 3.

A.1 Residual Income and Consumption Variances - Alternative Measures

Our benchmark empirical results on income and consumption variances are obtained by regressing the logarithm of disposable income on household characteristics including the sex and age of the household head, household size, education, province of residence, and ethnic minority status. This is done separately by year and for urban and rural households. The residual variance is then decomposed into transitory and permanent components.

In this appendix, we show residual variances without controls for education and gender of the household head. Figure A.1 shows variances of logs with controls for all other household characteristics mentioned above but without education and gender. Figure A.2 does the same but includes the education control and only omits gender. Figure A.3 shows the outcome when all controls besides education are present. Panels A - C in Table A.1 provide the numbers used to plot Figures A.1 to Figures A.3. Our results remain robust regardless of the controls on education or sex.

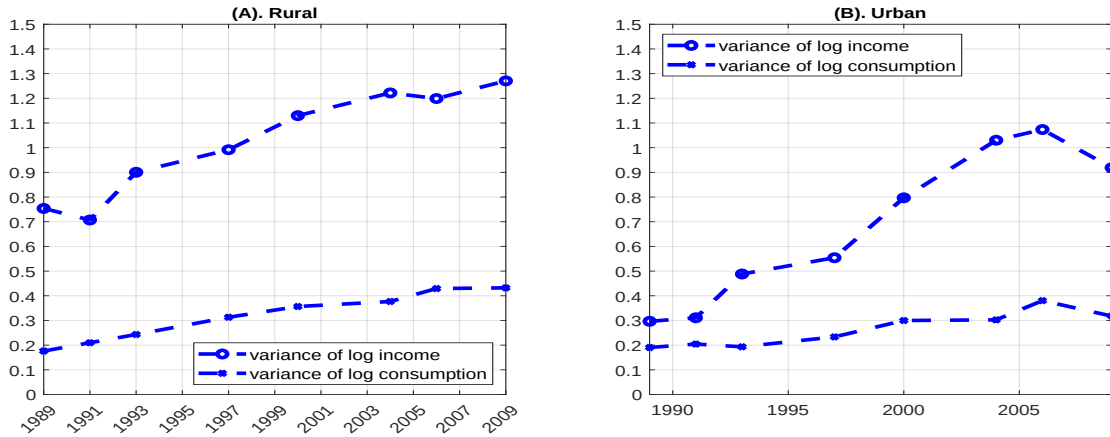


Figure A.1. Variance of Logarithm of Income and Consumption for Rural (left) and Urban (right) Households without Controls for Education or Gender.

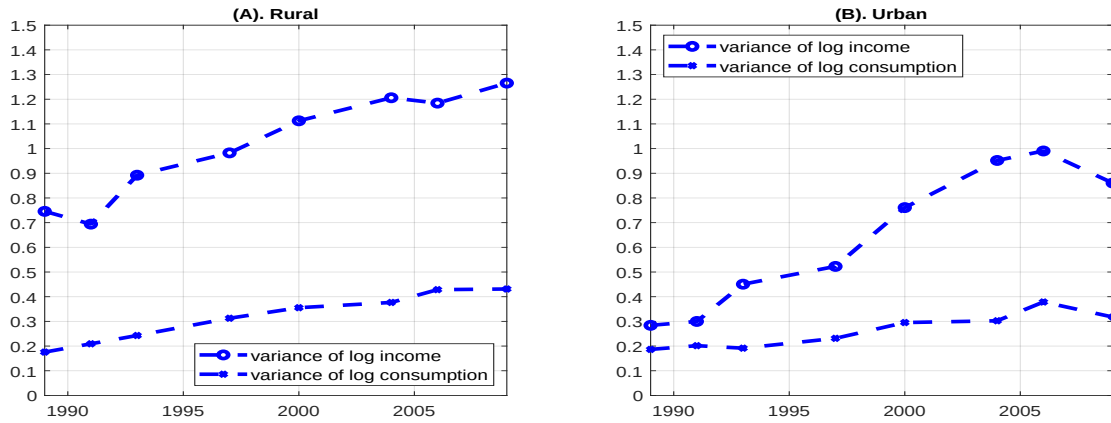


Figure A.2. Variance of Logarithm of Income and Consumption for Rural (left) and Urban (right) Households without Controls for Gender.

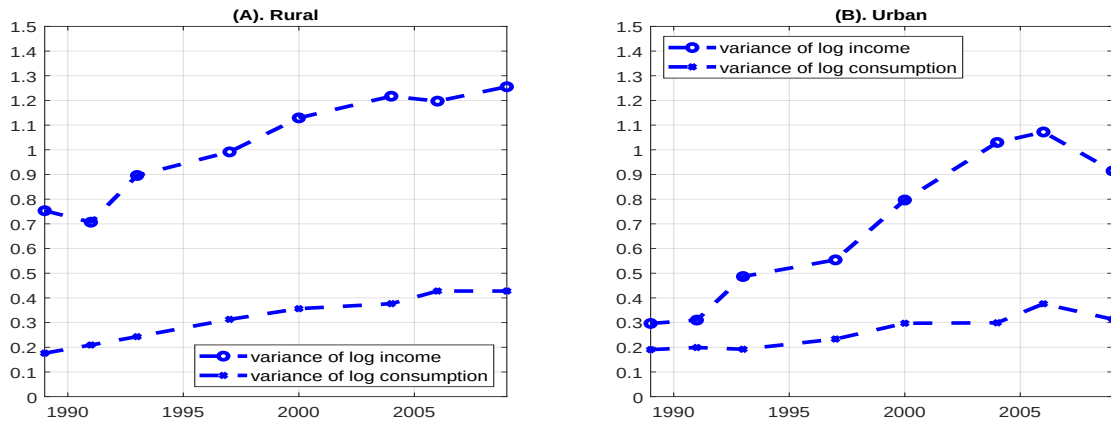


Figure A.3. Variance of Logarithm of Income and Consumption for Rural (left) and Urban (right) Households without Controls for Education.

Table A.1. Residual Income and Consumption Variances and Alternative Controls for Household Characteristics

Wave	var c rural	var c urban	var y rural	var y urban
<i>Panel A. Without control for sex or education</i>				
1989	0.176	0.190	0.754	0.297
1991	0.210	0.204	0.707	0.311
1993	0.243	0.193	0.900	0.488
1997	0.313	0.233	0.992	0.554
2000	0.356	0.300	1.130	0.797
2004	0.377	0.302	1.222	1.030
2006	0.429	0.380	1.199	1.073
2009	0.432	0.318	1.271	0.919
<i>Panel B. Control for sex but not education</i>				
1989	0.176	0.190	0.753	0.296
1991	0.209	0.199	0.707	0.310
1993	0.243	0.192	0.896	0.487
1997	0.313	0.233	0.991	0.554
2000	0.356	0.297	1.129	0.797
2004	0.377	0.299	1.217	1.030
2006	0.428	0.376	1.197	1.072
2009	0.428	0.313	1.255	0.914
<i>Panel C. Control for education but not sex</i>				
1989	0.176	0.186	0.746	0.284
1991	0.209	0.202	0.694	0.300
1993	0.243	0.191	0.892	0.451
1997	0.313	0.231	0.983	0.523
2000	0.355	0.295	1.113	0.761
2004	0.377	0.302	1.206	0.952
2006	0.429	0.379	1.184	0.990
2009	0.431	0.318	1.265	0.860
<i>Panel D. With all benchmark controls (including controls for both sex and education)</i>				
1989	0.175	0.186	0.744	0.284
1991	0.209	0.198	0.693	0.300
1993	0.243	0.191	0.885	0.451
1997	0.313	0.231	0.983	0.522
2000	0.355	0.293	1.110	0.760
2004	0.376	0.299	1.200	0.952
2006	0.427	0.374	1.183	0.988
2009	0.427	0.313	1.251	0.855

A.2 Income Shocks Estimation

To obtain minimum distance estimates of transitory and permanent income risks, we first obtain the adult-equivalent income by dividing household disposable income by the number of working age adults in the household. We then take the logarithm and regress it separately for urban and rural households and by year on a set of dummies on sex of household head, household head age, education, province of residence, and ethnic minority status. The residuals are used to estimate the permanent and transitory variances. Let y_t be the residual income:

$$\begin{aligned} y_t &= w_t + v_t \\ w_t &= w_{t-1} + \theta_t, \end{aligned}$$

where w_t represent the permanent components of income, θ_t is the shock to the persistent component of income (permanent shock) with mean zero and variance $\sigma_{\theta_t}^2$, and v_t is the transitory shock with mean zero and variance $\sigma_{v_t}^2$. Permanent and transitory shocks are mutually orthogonal and i.i.d. across household, time, and age.

Following the definition, the log income residual ($T > t$) can be rewritten as:

$$\begin{aligned} y_t &= w_{t-1} + \theta_t + v_t \\ y_{t+1} &= w_t + \theta_{t+1} + v_{t+1} \\ y_{t+2} &= w_{t+1} + \theta_{t+2} + v_{t+2} \\ &\dots \\ y_T &= w_{T-1} + \theta_T + v_T \\ \Delta y_{T-t} &= y_T - y_t \\ &= (\theta_{t+1} + \theta_{t+2} + \dots + \theta_T) - v_t + v_T, \end{aligned}$$

The moment conditions are expressed as: for any $T_1 > t$, $T_2 < t$,

$$\text{var}(\Delta y_{T-t}) = \sigma_{\theta_{t+1}}^2 + \sigma_{\theta_{t+2}}^2 + \dots + \sigma_{\theta_T}^2 + \sigma_{v_t}^2 + \sigma_{v_T}^2$$

$$\begin{aligned} \text{cov}(\Delta y_{T_1-t}, \Delta y_{t-T_2}) &= \text{cov}(y_{T_1} - y_t, y_t - y_{T_2}) \\ &= \text{cov}(\theta_{t+1} + \theta_{t+2} + \dots + \theta_{T_1} - v_t + v_{T_1}, \\ &\quad \theta_{T_2+1} + \theta_{T_2+2} + \dots + \theta_t - v_{T_2} + v_t) \\ &= \text{cov}(-v_t, v_t) \\ &= -\sigma_{v_t}^2. \end{aligned}$$

The variances and covariances on the left-hand side are estimated from the data, then the parameters for the transitory and permanent variances are chosen to minimize the distance between the model predictions and their data counterparts.

The permanent income shocks are assumed to be constant between two survey years. There are seven variance conditions estimated from the data, $\text{var}(\Delta y_{T-t})$ for $t = 1991, 1993, 1997, 2000, 2004, 2006, 2009$; and six covariance conditions estimated from the data, $\text{cov}(\Delta y_{T_1-t}, \Delta y_{t-T_2})$ for $t = 1991, 1993, 1997, 2000, 2004, 2006$. The five parameters of permanent variance are estimated for $t = 1993, 1997, 2000, 2004, 2006$; and the six parameters of transitory variance are estimated for $t = 1991, 1993, 1997, 2000, 2004, 2006$.

B Calibration

In this appendix, we provide additional materials related to Section 5.

B.1 Labor Participation Ratio by Age and Wave

We use participants' answers to the question "Are you presently working/employed?" in the CHNS survey questionnaire to define labor force participation. Figure B.1 shows a steadily declining trend in labor force participation rate after age 50 in both rural and urban samples. This pattern has important implications for our treatment of old households in the theoretical model, especially for rural households.

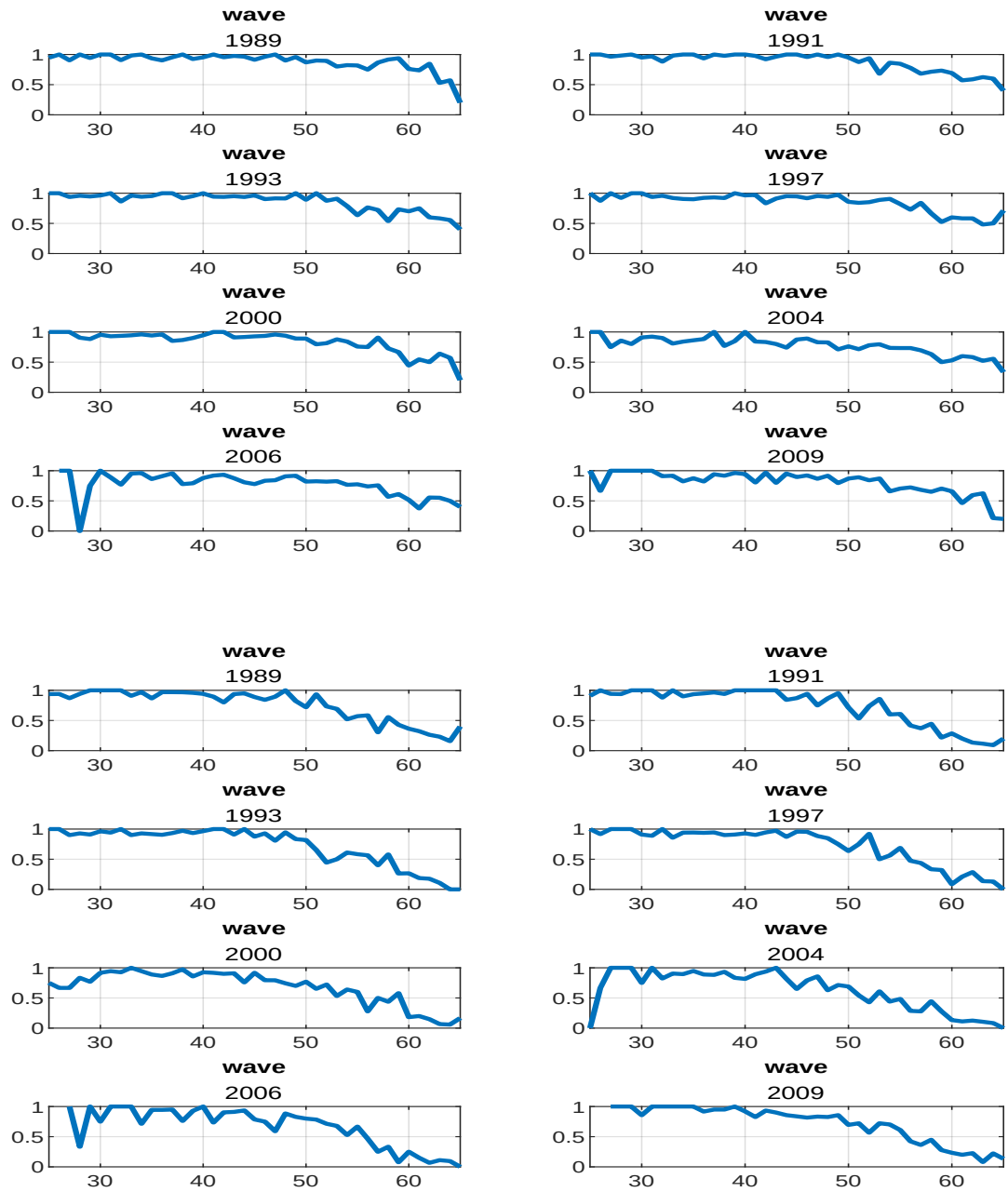


Figure B.1. Labor Participation Rate by Age and Wave

Notes: The rural sample is shown in the upper panel, and the urban in the lower. Data source: CHNS.

B.2 Urban-Rural Age-Dependent Survival Rate

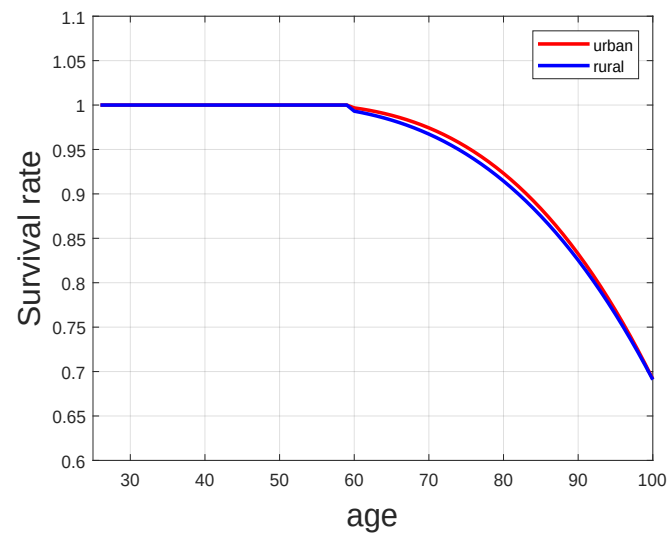


Figure B.2. Urban and Rural Old Age Survival Rates

Data source: [Zimmer et al. \(2007\)](#). Lines have been fit using cubic regression of survival rate on age.

B.3 Estimation of Deterministic Income Growth Rate by Age

The deterministic life-cycle income growth rate before retirement is estimated by regressing the log of household disposable income on a set of cohort dummies, household size, and a fourth order polynomial of the age of the household head. The estimation method can be expressed as follows:

$$Y = X\beta + \epsilon$$

where Y denotes log of household disposable income and X is a vector of cohort dummies, household size, and a fourth order polynomial in the age of the household head. Figure B.3 plots the estimated marginal effect of age on household disposable income before age 58.

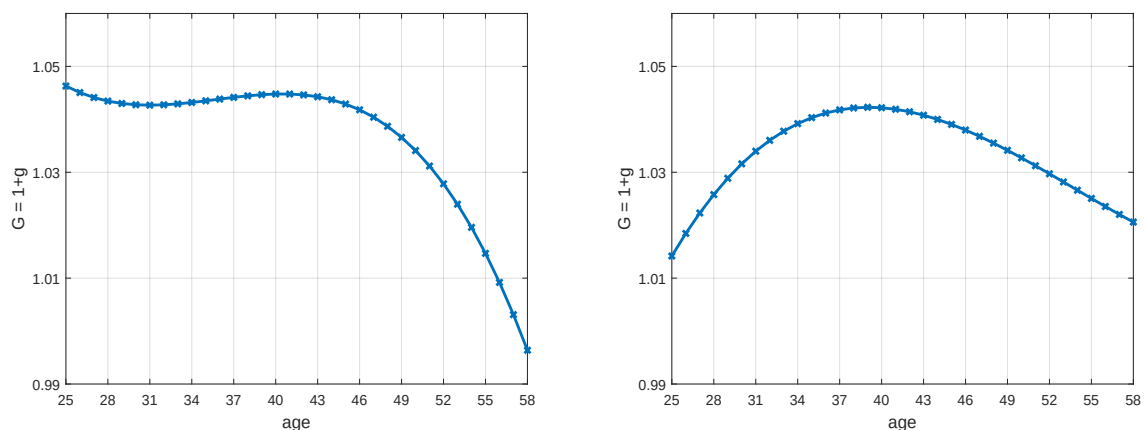


Figure B.3. Gross Income Growth Rate by Age

Notes: The left panel shows the rural sample, and the right panel shows the urban sample. Income growth rate at each age is calculated as the marginal effect of age on household disposable income plus 1.

C Computational Procedures

In this appendix, we describe the solution algorithm. We iterate over an approximation of the general Euler equation to find the optimal consumption at each age. The Euler equation can be found as the following:

$$\begin{aligned} \max \left\{ \sum_{t=25}^{100} \beta^t \left(\prod_{j=25}^t s_j \right) E_t[\mu(C_t)] \right\}, \quad \mu(C_t) = \frac{C_t^{1-\gamma}}{1-\gamma} \\ \text{s.t.} \quad A_{t+1} = (1+r)(A_t + Y_t - C_t), \\ A_t \geq 0, \quad \forall t. \end{aligned}$$

This problem can be expressed in a Lagrangian function as:

$$\mathcal{L} = \sum_{t=25}^{100} \beta^t \left(\prod_{j=25}^t s_j \right) E_t[\mu(C_t)] + \lambda_t [A_{t+1} - (1+r)(A_t + Y_t - C_t)]. \quad (\text{C.1})$$

Take the first order condition over C_t and C_{t+1} :

$$\frac{\partial \mathcal{L}_t}{\partial C_t} = \beta^t \left(\prod_{j=25}^t s_j \right) \mu'(C_t) + \lambda_t (1+r), \quad (\text{C.2})$$

$$\frac{\partial \mathcal{L}_t}{\partial C_{t+1}} = \beta^{t+1} \left(\prod_{j=25}^{t+1} s_j \right) E_t[\mu'(C_{t+1})] + \lambda_{t+1} (1+r). \quad (\text{C.3})$$

$$(\text{C.4})$$

Let $\frac{\partial \mathcal{L}_t}{\partial C_t} = 0$ and $\frac{\partial \mathcal{L}_t}{\partial C_{t+1}} = 0$:

$$-\lambda_t (1+r) = \beta^t \left(\prod_{j=25}^t s_j \right) \mu'(C_t) \quad (\text{C.5})$$

$$\lambda_t = -\beta^{t+1} \left(\prod_{j=25}^{t+1} s_j \right) E_t[\mu'(C_{t+1})] \quad (\text{C.6})$$

Plug equation C.5 into C.4:

$$\beta^t \left(\prod_{j=25}^t s_j \right) \mu'(C_t) = -(1+r) \beta^{t+1} \left(\prod_{j=25}^{t+1} s_j \right) E_t[\mu'(C_{t+1})] \quad (\text{C.7})$$

Simplifying C.6, we get:

$$\mu'(C_t) = \beta(1+r)s_{t+1}E_t[\mu'(C_{t+1})] \quad (\text{C.8})$$

Using $\mu'(C_t) = C_t^{-\gamma}$, equation C.7 can be further simplified to:

$$C_t^{-\gamma} = \beta(1+r)s_{t+1}E_t[C_{t+1}^{-\gamma}] \quad (\text{C.9})$$

Equation (C.8) is the Euler equation. Divide variables in equation (C.8) by W_t :

$$c_t^{-\gamma} = \beta(1+r)s_{t+1}E_t[(c_{t+1}G_{t+1}\theta_{t+1})^{-\gamma}] \quad (\text{C.10})$$

Given the evolution of the real assets, $A_{t+1} = (1+r)(A_t + Y_t - C_t)$, define $X_t = A_t + Y_t$ as the “cash on hand,” the resource available for consumption at time period t . The evolution of “cash on hand” is given by:

$$X_{t+1} = (1+r)(X_t - C_t) + Y_{t+1} \quad (\text{C.11})$$

Dividing through all the variables by the level of permanent income (W_{t+1}), and then using lower case variables as the upper case variables divided by permanent income, we arrive at:

$$x_{t+1} = \frac{1+r}{G_{t+1}\theta_{t+1}}(x_t - c_t) + V_{t+1} \quad (\text{C.12})$$

At any time t , c_t is a function of x_t , $c_t(x_t)$. Rewrite C.9 to express the Euler equation in the general form:

$$1 = \beta(1+r)s_{t+1}E_t \left[\left\{ c_{t+1} \left(\frac{1+r}{G_{t+1}\theta_{t+1}}(x_t - c_t) + V_{t+1} \right) G_{t+1}\theta_{t+1}/c_t \right\}^{-\gamma} \right] \quad (\text{C.13})$$

The model is solved by backwards induction on the approximation of the Euler equation starting from the last period. The optimal solution for consumption in period T is to consume all available resources, $c_T^*(x_T) = x_T$. Construct the grid of possible values of state space as a triple exponential. Given the 100 grid points of the state space $x_1, x_2, x_3, \dots, x_{100}$, time discount factor β , interest rate r , age-dependent survival rate s_i , gross permanent income growth rate G_t , and the distributions of θ_t and v_t , the iteration

is carried out as follows:

1. Start from the last period T and approximate the Euler equation.
2. For any x_i , invert the approximated Euler equation to find the optimal value of c_{T-1}^* . The numerical optimal consumption is constructed by linear interpolation between grid points of the state space.
3. The no-borrowing budget constraint implies that the next period’s “cash on hand,” x_{T-1} , equals $c_{T-1}^* + x_T$. Note here, unlike the standard approach, the grid points of the state space are not predetermined for each period. Rather, the grid over the state space in period t is endogenously dependent on the end-of-period “cash on hand” value x_{t+1} (the endogenous grid point method).
4. Given the new grids of the state space, repeat the first three steps until the first period is reached.
5. Repeat steps 1 - 4 for each combination of income parameters. That is, for urban households, given the baseline parameters in 1993, shocks to income variance in years 1994, 1998, 2000, 2001, and 2005, shocks to income growth rate and pension replacement rate in year 1998, we solve six policy functions. For rural households, given the baseline parameters in 1993, shocks to income variance in years 1994, 1998, 2000, 2001, and 2005, shocks to income growth rate in 1998, and pension shock in year 2003, we solve seven policy functions.
6. Simulate life-cycle profiles for six cohorts aged between 25 and 55 years in 1993 with 5-year intervals. Each cohort consists of 10,000 households. Simulations are carried out separately for urban and rural samples.

D Household Saving Rate

In this appendix, we discuss how we compute the household saving rate presented in Section 7 from the Chinese Household Income Projects (CHIPs). The CHIPs is the only publicly available dataset that contains detailed Chinese household income and consumption information necessary for calculating saving rates for the period of interest.

The CHIPs was first launched in 1988, and the most recent available survey was in 2013. This paper only uses three years of data (1995, 2002 and 2007) from the project,

mainly due to the quality of the data.¹⁴ It conglomerates separate survey data studying rural and urban populations into one central format. Specifically, the samples are subsets of larger nationally-representative samples (67,186 rural households and 34,945 urban households) drawn by National Bureau of Statistics (NBS) in the Urban Household Survey (UHS) and Rural Household Survey (RHS). The samples are drawn by using an income based, equidistant random sampling method.

The CHIPs is primarily designed to follow the levels and distribution of Chinese household income. It also provides detailed information on household-level consumption expenditures and family demographic characteristics for selected years. The measure of household disposable income is composed of labor income, operating income, property income, capital income, agricultural income, transfer income (both public and private), and income from household sideline production. The consumption expenditures includes a broad range of categories: food, clothing and footwear, home appliances, health expenditure, transportation and communication, education, culture, recreation, housing and services, utility bills, and sundries. All values are denoted in 2007 RMB.

Table D.1 presents the summary statistics of urban-rural household saving rates and household characteristics in the CHIPs data. The household saving rate is computed as 1 - consumption/disposable income. Rural households are these households whose head is a rural *hukou* holder, urban households are the households whose head is a urban *hukou* holder. Samples are restricted to households with head aged between 25 and 60.

As is clear in the data, the average household income and consumption expenditures for both urban and rural households increased sharply in the period of interest (1995-2007), accompanied by equally fast-rising variances. A steeper increase in household income was observed in urban households, with household disposable income more than tripling from 18,241 RMB (\$2,682) in 1995 to 60,170 RMB (\$8,848) in 2007.¹⁵ Household consumption, on the other hand, increased from 12,230 RMB (\$1,798) to 36,672 RMB (\$5,222). Income and consumption more than doubled for rural households, with average income increasing from 8,588 RMB (\$1,257) in 1995 to 19,968 RMB (\$2,923) in

¹⁴We drop 1988's survey data as some key income and consumption expenditures information is missing. The 1999 urban data is also dropped, as they only surveyed urban households in that year. We focus on the years before 2013. The CHIPs also added surveys specifically for rural-urban migrants since 2002. Due to the complex features of that sample, we leave it for future research.

¹⁵The dollar incomes in the parentheses are equal to the Yuan amount divided by an exchange rate of 6.83.

1997, consumption increased from 4,044 RMB (\$592) to 14,772 RMB (\$2,162). The average household saving rates, nevertheless, displayed opposite trends in urban and rural samples. That is, the aggregate urban household saving rate increased from 12 percent in 1995 to 34 percent in 2007, at the same time rural households saw a decrease from 29 percent to 17 percent. This demonstrates an over 20 percentage point increase in the urban household saving rate, while the rural household saving rate saw a 10 percentage point decline.

Table D.1. Summary Statistics, CHIPs

Wave	Observations (households)	Head Age	Household Size	Disposable Income (In 2007 RMB)		Consumption (In 2007 RMB)		Household Saving Rate		At least some high school		At least one member employed by SOE	
				Mean	Std.dev.	Mean	Std.dev.	Mean	Std.dev.	Mean	Std.dev.	Mean	Std.dev.
Urban													
1995	5,672	43	3.2	18,241	9,916	12,230	7,064	0.12	0.24	0.59	0.49	0.35	0.48
2002	5,582	44	3.0	25,078	15,799	18,875	12,268	0.21	0.28	0.63	0.48	0.15	0.36
2007	3,732	44	3.0	60,170	43,656	35,672	24,557	0.34	0.26	0.66	0.47	0.28	0.50
Rural													
1995	6,725	42	4.2	8,106	5,237	4,744	3,843	0.42	0.65	0.09	0.29	0.06	0.24
2002	7,833	45	4.1	11,963	8,701	7,540	5,669	0.20	0.40	0.13	0.34	0.06	0.23
2007	6,616	48	4.0	19,968	14,049	14,772	12,907	0.17	0.52	0.13	0.33	-	-

Notes: Saving rate=1-consumption/disposable income. Income and consumption are household-level income and consumption, inflated by provincial CPI, weighted by sample province weights, expressed in 2007 RMB. '-' not applicable. Data source: CHIPs (1995, 2002, 2007).